

**Supplementary materials for the paper “On Testing Kronecker Product
Structure in Tensor Factor Models” - Model identification, and proof of all
theorems and auxiliary results**

S1 Appendix: Identification

This section concerns the identification of the model in Definition 2, following a discussion on Definition 1. First, consider Definition 2.1 so that in (3.1), we have $\mathbf{A}_{\text{reshape}, K-\ell+1} \in \mathcal{K}_{d_{a_1} \times \dots \times d_{a_\ell}}$. Given a general $\mathcal{A} = \{a_1, \dots, a_\ell\}$, the ordered set of matrices $\{\mathbf{A}_j\}_{j \in [\kappa]}$ decomposing (as in Definition 1) $\mathbf{A}_{\text{reshape}, K-\ell+1}$ might not be unique. For instance, suppose $K = 2$, $d_1 = d_2$, $\mathcal{A} = \{1, 2\}$ (hence $\mathbf{A}_{\text{reshape}, K-\ell+1} \equiv \mathbf{A}_{\text{reshape}, 1}$ has d_1^2 rows) and let $\mathbf{A}_{\text{reshape}, 1} = d_1^{-1/4} \mathbf{1}_{d_1^2}$, then we have

$$\mathbf{A}_{\text{reshape}, 1} = \underbrace{(d_1^{-1/4} \mathbf{1}_{d_1})}_{\mathbf{A}_1} \otimes \underbrace{\mathbf{1}_{d_1}}_{\mathbf{A}_2} = \underbrace{\mathbf{1}_{d_1}}_{\check{\mathbf{A}}_1} \otimes \underbrace{(d_1^{-1/4} \mathbf{1}_{d_1})}_{\check{\mathbf{A}}_2}, \quad (\text{S1.1})$$

where it is clear that $\|\mathbf{A}_{1,1}\|^2 \asymp d_1^{1/2}$ and $\|\check{\mathbf{A}}_{1,1}\|^2 \asymp d_1$. Such defeat can be rectified by allocating the “factor strength” in $\mathbf{A}_{\text{reshape}, K-\ell+1}$ to each mode in $\mathcal{A}$; see the following Assumption (S1) as one example.

(S1) For (3.2) such that $\mathbf{A}_{\text{reshape}, K-\ell+1} = \otimes_{i \in \mathcal{A}} \mathbf{A}_i$ for a given $\mathcal{A}$, we assume that for any $i \in \mathcal{A}$,

$$\frac{\|\mathbf{A}_i\|_F^2}{r_i d_i} = \left(\frac{\|\mathbf{A}_{\text{reshape}, K-\ell+1}\|_F^2}{\prod_{i \in \mathcal{A}} r_i d_i} \right)^{1/|\mathcal{A}|}.$$

The issue of indeterminacy in the factor strength is fixed by (S1) which has same spirit of Assumption (IC) in Chen and Lam (2024). Its heuristic is to allocate the factor strength in $\mathbf{A}_{\text{reshape}, K-\ell+1}$ to each mode according to their number of factors and dimensions. Note that Assumption (S1) holds automatically if all factors are pervasive. Now recall the example (S1.1), $\mathbf{A}_1 = \mathbf{A}_2 = d_1^{-1/8} \mathbf{1}_{d_1}$ are identified by (S1). Note that the discussion above is only for completeness and would not influence our testing problem.

Next, we identify models in Definition 2 and Theorem 1. For (3.1) and (3.2), we state below the two assumptions (L1') and (L2'), which are (notation-wise) general versions of (L1) and (L2), respectively.

(L1') (Factor strength in $\mathbf{A}_{\text{reshape}, j}$) For each $j \in [K - \ell]$, we assume that $\mathbf{A}_{\text{reshape}, j}$ in (3.1) is of full rank and as $I_j \rightarrow \infty$,

$$\mathbf{Z}_{\text{reshape}, j}^{-1/2} \mathbf{A}'_{\text{reshape}, j} \mathbf{A}_{\text{reshape}, j} \mathbf{Z}_{\text{reshape}, j}^{-1/2} \rightarrow \boldsymbol{\Sigma}_{\text{reshape}, A, j}, \quad (\text{S1.2})$$

where $\boldsymbol{\Sigma}_{\text{reshape}, A, j}$ is positive definite with all eigenvalues bounded away from 0 and infinity, and $\mathbf{Z}_{\text{reshape}, j}$ is a diagonal matrix with $(\mathbf{Z}_{\text{reshape}, j})_{hh} \asymp I_j^{\delta_{\text{reshape}, j, h}}$ for $h \in [\pi_j]$ and the ordered factor

strengths $1/2 < \delta_{\text{reshape},j,\pi_j} \leq \dots \leq \delta_{\text{reshape},j,2} \leq \delta_{\text{reshape},j,1} \leq 1$.

We assume that $\mathbf{A}_{\text{reshape},j}$ for $j = K - \ell + 1$ also has the above form, except that only the maximum and minimum factor strengths are ordered, i.e., $1/2 < \delta_{\text{reshape},j,\pi_j} \leq \delta_{\text{reshape},j,h} \leq \delta_{\text{reshape},j,1} \leq 1$ for any $h \in [\pi_j]$.

(L2') (Factor strength in $\mathbf{A}_i$) For (3.2) with a given $\mathcal{A}$, we assume that for each $i \in \mathcal{A}$, $\mathbf{A}_i$ is of full rank and as $d_i \rightarrow \infty$,

$$\mathbf{Z}_i^{-1/2} \mathbf{A}_i' \mathbf{A}_i \mathbf{Z}_i^{-1/2} \rightarrow \Sigma_{A,i}, \quad (\text{S1.3})$$

where $\Sigma_{A,i}$ is positive definite with all eigenvalues bounded away from 0 and infinity, and $\mathbf{Z}_i$ is a diagonal matrix with $(\mathbf{Z}_i)_{hh} \asymp d_i^{\delta_{i,h}}$ for $h \in [r_i]$ and the ordered factor strengths $1/2 < \delta_{i,r_i} \leq \dots \leq \delta_{i,2} \leq \delta_{i,1} \leq 1$.

Theorem 1 presents the identification of the model (3.1) both in general and with Kronecker product structure (equivalently (3.2) according to Theorem 1). Its proof is given directly after the statement.

Theorem 1 (Identification) *Let Assumption (F1) and (L1') hold, and $\mathcal{A}$ is given. Then the factor structure in (3.1) is asymptotically identified up to some invertible matrix $\mathbf{M}_j \in \mathbb{R}^{\pi_j \times \pi_j}$ such that the following sets of factor structure are equivalent,*

$$\left(\mathcal{F}_{\text{reshape},t}, \{ \mathbf{A}_{\text{reshape},j} \}_{j \in [K-\ell+1]} \right) = \left(\mathcal{F}_{\text{reshape},t} \times_{j=1}^{K-\ell+1} \mathbf{M}_j^{-1}, \{ \mathbf{A}_{\text{reshape},j} \mathbf{M}_j \}_{j \in [K-\ell+1]} \right).$$

Let Assumption (L2') further holds. With a Kronecker product structure on (3.1), we have (3.2) where for each $k \in [K]$, $\mathbf{A}_k$ has unique rank and the factor structure in (3.2) is asymptotically identified up to some invertible matrices.

Proof of Theorem 1. Consider first (3.1). Let $(\ddot{\mathcal{F}}_{\text{reshape},t}, \{ \ddot{\mathbf{A}}_{\text{reshape},j} \}_{j \in [K-\ell+1]})$ be another set of parameters such that $\ddot{\mathcal{F}}_{\text{reshape},t} \times_{j=1}^{K-\ell+1} \ddot{\mathbf{A}}_{\text{reshape},j} = \mathcal{F}_{\text{reshape},t} \times_{j=1}^{K-\ell+1} \mathbf{A}_{\text{reshape},j}$. Define

$$\mathbf{A}_{\text{reshape},-j} := \mathbf{A}_{\text{reshape},K-\ell+1} \otimes \dots \otimes \mathbf{A}_{\text{reshape},j+1} \otimes \mathbf{A}_{\text{reshape},j-1} \otimes \dots \otimes \mathbf{A}_{\text{reshape},1}. \quad (\text{S1.4})$$

Define $\ddot{\mathbf{A}}_{\text{reshape},-j}$ similarly. Without loss of generality, for any $j \in [K - \ell + 1]$ we write

$$\ddot{\mathbf{A}}_{\text{reshape},j} = \mathbf{A}_{\text{reshape},j} \mathbf{M}_{\text{reshape},j} + \mathbf{\Gamma}_{\text{reshape},j}, \quad \text{where } \mathbf{\Gamma}_{\text{reshape},j}' \mathbf{A}_{\text{reshape},j} = \mathbf{0}, \quad (\text{S1.5})$$

with $\mathbf{M}_{\text{reshape},j} \in \mathbb{R}^{\pi_j \times \pi_j}$ and $\mathbf{\Gamma}_{\text{reshape},j} \in \mathbb{R}^{I_j \times \pi_j}$, but can have zero columns. Then

$$\begin{aligned} \mathbf{0} &= \mathbf{\Gamma}'_{\text{reshape},j} \mathbf{A}_{\text{reshape},j} \mathbf{F}_{\text{reshape},t,(j)} \mathbf{A}'_{\text{reshape},-j} \ddot{\mathbf{A}}_{\text{reshape},-j} \\ &= \mathbf{\Gamma}'_{\text{reshape},j} \ddot{\mathbf{A}}_{\text{reshape},j} \ddot{\mathbf{F}}_{\text{reshape},t,(j)} \ddot{\mathbf{A}}'_{\text{reshape},-j} \ddot{\mathbf{A}}_{\text{reshape},-j} \\ &= \mathbf{\Gamma}'_{\text{reshape},j} \mathbf{\Gamma}_{\text{reshape},j} \ddot{\mathbf{F}}_{\text{reshape},t,(j)} \ddot{\mathbf{A}}'_{\text{reshape},-j} \ddot{\mathbf{A}}_{\text{reshape},-j}, \end{aligned}$$

which can only be true in general if $\mathbf{\Gamma}_{\text{reshape},j} = \mathbf{0}$ since $\ddot{\mathbf{F}}_{\text{reshape},t,(j)}$ is random by Assumption (F1) and $\ddot{\mathbf{A}}'_{\text{reshape},-j} \ddot{\mathbf{A}}_{\text{reshape},-j}$ converges to some full rank matrix by (L1'). Hence $\mathbf{M}_{\text{reshape},j}$ has full rank, and $\ddot{\mathbf{A}}_{\text{reshape},j}$ and $\mathbf{A}_{\text{reshape},j}$ share the same column space. $\mathcal{F}_{\text{reshape},t}$ is identified once $\{\mathbf{A}_{\text{reshape},j}\}_{j \in [K-\ell+1]}$ is given correspondingly.

Suppose (3.1) has a Kronecker product structure and consider (3.2). By an argument similar to (S1.5) but over $k \in [K]$ (omitted), each matrix $\mathbf{A}_k$ for $k \in [K]$ has a unique rank and is identified up to some invertible matrix using (L2'). Hence $\mathcal{F}_t$ is also identified, which completes the proof of the theorem. $\square$

S2 Appendix: Proof of theorems and lemmas

A high-level summary of proofs:

The design of (3.9) and (3.10) over all divisor combinations of r_V is due to pseudo-ranks from mode- v to mode- K of $\mathcal{Y}_t$ under H_1 in (3.4). On the other hand, under H_0 , there must be one $m \in [|\mathcal{R}|]$ such that $(\pi_{m,1}, \dots, \pi_{m,K-v+1}) = (r_v, \dots, r_K)$. Hence we consider throughout the proof that $\{r_k\}_{k \in [K]}$ in (3.5) (hence under H_0) is correctly specified. Hence, we simplify the notations $\hat{\mathbf{Q}}_{m,i}$, $\hat{\mathcal{C}}_{m,t}$ and $\hat{\mathcal{E}}_{m,t}$ as $\hat{\mathbf{Q}}_i$, $\hat{\mathcal{C}}_t$ and $\hat{\mathcal{E}}_t$, respectively. Whenever (L2) is assumed, we are implicitly considering H_0 in (3.4), where we also define

$$\begin{aligned} \mathbf{Q}_{-k} &:= \mathbf{Q}_K \otimes \dots \otimes \mathbf{Q}_{k+1} \otimes \mathbf{Q}_{k-1} \otimes \dots \otimes \mathbf{Q}_1 \quad \text{for } k \in [K], \\ \mathcal{F}_{Z,t} &:= \mathcal{F}_t \times_{k=1}^K \mathbf{Z}_k^{1/2} \quad \text{with } \mathcal{F}_t \text{ from (4.5)}. \end{aligned}$$

Theorem 1 reveals the importance of the reshape operator and is the key to formalise the testing problem. Lemma 1 to Lemma 4 serve as technical steps. The steps of all other proofs are as follows.

1. Under H_0 , consider (3.5). We first derive the rate of convergence for $\hat{\mathbf{Q}}_k$ as an estimator of $\mathbf{Q}_k$ for each $k \in [K]$ (Lemma 5). Then the rates for the corresponding core factor and hence the common component can also be obtained (Lemma 6).
2. Under H_0 , consider (3.3). We derive the rates of convergence for $\{\tilde{\mathbf{Q}}_j\}_{j \in [v-1]}$ and $\tilde{\mathbf{Q}}_V$ as estimators of

$\{\mathbf{Q}_j\}_{j \in [v-1]}$ and $\mathbf{Q}_V$, respectively. Similar to step 1, the rate for the common component is obtained. See Lemma 7.

3. With steps 1 and 2, we then show that $\sum_{i=1}^{d_{k^*}} \widehat{E}_{m,t,(k^*),ij}^2$ and $\sum_{i=1}^{d_{k^*}} \widetilde{E}_{t,(k^*),ij}^2$ under H_0 have the same distribution asymptotically (Theorem 2).
4. With step 3, the test statistic can be constructed with theoretical support (Theorem 3).

S2.1 Lemmas and proofs

By the interplay between the core factor and the noise series in Assumptions (F1), (E1) and (E2), we state below Lemma 1 which is direct from Proposition 1.1 and 1.2 of Cen and Lam (2024).

Lemma 1 *Let Assumptions (F1), (E1) and (E2) hold. Then*

1. *(Weak correlation of noise $\mathcal{E}_t$ across different modes and time) there exists some positive constant $C < \infty$ so that for any $t \in [T]$, $k \in [K]$, $i_k, j \in [d_k]$, $h \in [d_{-k}]$, we have $\mathbb{E}\mathcal{E}_{t,i_1,\dots,i_K} = 0$, $\mathbb{E}\mathcal{E}_{t,i_1,\dots,i_K}^4 \leq C$, and*

$$\begin{aligned} \sum_{j=1}^{d_k} \sum_{l=1}^{d_{-k}} \left| \mathbb{E}[E_{t,(k),i_k h} E_{t,(k),jl}] \right| &\leq C, \\ \sum_{l=1}^{d_{-k}} \sum_{s=1}^T \left| \text{Cov}(E_{t,(k),i_k h} E_{t,(k),jh}, E_{s,(k),i_k l} E_{s,(k),jl}) \right| &\leq C. \end{aligned}$$

2. *(Weak dependence between factor $\mathcal{F}_t$ and noise $\mathcal{E}_t$) there exists some positive constant $C < \infty$ so that for any $k \in [K]$, $j \in [d_k]$, and any deterministic vectors $\mathbf{u} \in \mathbb{R}^{r_k}$ and $\mathbf{v} \in \mathbb{R}^{r_{-k}}$ with constant magnitudes, it holds for $\mathcal{F}_t$ in (3.5) that*

$$\mathbb{E} \left(\frac{1}{(d_{-k}T)^{1/2}} \sum_{h=1}^{d_{-k}} \sum_{t=1}^T E_{t,(k),jh} \mathbf{u}' \mathbf{F}_{t,(k)} \mathbf{v} \right)^2 \leq C.$$

3. *Statement 2 holds similarly for $\text{Reshape}(\mathcal{F}_t, \mathcal{A})$ and $\text{Reshape}(\mathcal{E}_t, \mathcal{A})$.*

Lemma 2 *Under Assumption (F1), with $\gamma_v := \prod_{j=1}^{v-1} r_j$, we have as $T \rightarrow \infty$,*

$$\frac{1}{T} \sum_{t=1}^T \mathbf{F}_{t,(k)} \mathbf{M} \mathbf{F}_{t,(k)}' \xrightarrow{p} \text{tr}(\mathbf{M}) \cdot \mathbf{I}_{r_k}, \quad (\text{S2.1})$$

$$\frac{1}{T} \sum_{t=1}^T \mathbf{F}_{\text{reshape},t,(j)} \mathbf{N} \mathbf{F}_{\text{reshape},t,(j)}' \xrightarrow{p} \text{tr}(\mathbf{N}) \cdot \mathbf{I}_{r_j}, \quad (\text{S2.2})$$

$$\frac{1}{T} \sum_{t=1}^T \mathbf{F}_{\text{reshape},t,(v)} \mathbf{W} \mathbf{F}_{\text{reshape},t,(v)}' \xrightarrow{p} \text{tr}(\mathbf{W}) \cdot \mathbf{I}_{r_V}, \quad (\text{S2.3})$$

where $\mathbf{F}_{t,(k)} \in \mathbb{R}^{r_k \times r_{-k}}$ for $k \in [K]$ represents the mode- k unfolding of $\mathcal{F}_t$ in (3.5), and $\mathbf{M}$ is any $r_{-k} \times r_{-k}$ matrix independent of $\{\mathcal{F}_t\}_{t \in [T]}$ with $\|\mathbf{M}\|_F$ bounded in probability; similarly for $\mathbf{F}_{\text{reshape},t,(j)} \in \mathbb{R}^{r_k \times (r_V \gamma_v / r_j)}$ for $j \in [v-1]$ in (3.3); similarly for $\mathbf{F}_{\text{reshape},t,(v)} \in \mathbb{R}^{r_V \times \gamma_v}$ in (3.3).

Proof of Lemma 2. It suffices to show (S2.1) and the other two follow similarly. With (F1), consider first

$$\begin{aligned} & \sum_{s=1}^T \left(\sum_{w \geq 0} a_{f,w} a_{f,w-(t-s)} \right) \left(\sum_{q \geq 0} a_{f,q} a_{f,q-(t-s)} \right) \\ & \leq \left(\sum_{s=1}^T \sum_{w \geq 0} |a_{f,w}| |a_{f,w-(t-s)}| \right) \left(\sum_{q \geq 0} |a_{f,q}| \right) \cdot \max_q |a_{f,q}| = O(1) \cdot \left(\sum_{w \geq 0} |a_{f,w}| \right)^2 = O(1), \end{aligned} \quad (\text{S2.4})$$

where the last two equality used Assumption (F1). Similarly, it holds that

$$\sum_{s=1}^T \sum_{w \geq 0} a_{f,w}^2 a_{f,w-(t-s)}^2 \leq \left(\sum_{w \geq 0} a_{f,w}^2 \right)^2 = O(1). \quad (\text{S2.5})$$

Now for $i \neq j \in [r_k]$, by Assumption (F1) we have

$$\mathbb{E} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{F}'_{t,(k),i} \cdot \mathbf{M} \mathbf{F}_{t,(k),j} \right) = \mathbb{E} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{F}'_{t,(k),i} \right) \mathbb{E} \left(\mathbf{M} \mathbf{F}_{t,(k),j} \right) = 0.$$

For $i = j \in [r_k]$, with $\mathcal{X}_{f,t}$ from (4.5), we have

$$\begin{aligned} & \mathbb{E} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{F}'_{t,(k),i} \cdot \mathbf{M} \mathbf{F}_{t,(k),i} \right) = \frac{1}{T} \sum_{t=1}^T \sum_{w \geq 0} \sum_{q \geq 0} a_{f,w} a_{f,q} \mathbb{E} \left(\mathbf{X}'_{f,t-w,(k),i} \cdot \mathbf{M} \mathbf{X}_{f,t-q,(k),i} \right) \\ & = \frac{1}{T} \sum_{t=1}^T \sum_{w \geq 0} \sum_{q \geq 0} a_{f,w} a_{f,q} \text{tr}(\mathbf{M}) \mathbb{1}\{w = q\} = \text{tr}(\mathbf{M}) \sum_{w \geq 0} a_{f,w}^2 = \text{tr}(\mathbf{M}), \end{aligned}$$

so that $\mathbb{E} \left(T^{-1} \sum_{t=1}^T \mathbf{F}_{t,(k)} \mathbf{M} \mathbf{F}'_{t,(k)} \right) = \text{tr}(\mathbf{M}) \cdot \mathbf{I}_{r_k}$. Finally, consider any $i, j \in [r_k]$,

$$\begin{aligned} & \text{Var} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{F}'_{t,(k),i} \cdot \mathbf{M} \mathbf{F}_{t,(k),j} \right) = \text{Var} \left(\frac{1}{T} \sum_{t=1}^T \sum_{l=1}^{r_{-k}} \sum_{v=1}^{r_{-k}} F_{t,(k),il} M_{lv} F_{t,(k),jv} \right) \\ & = \frac{1}{T^2} \text{Cov} \left(\sum_{t=1}^T \sum_{l=1}^{r_{-k}} \sum_{v=1}^{r_{-k}} \sum_{w \geq 0} \sum_{q \geq 0} a_{f,w} a_{f,q} X_{f,t-w,(k),il} M_{lv} X_{f,t-q,(k),jv}, \right. \\ & \quad \left. \sum_{s=1}^T \sum_{g=1}^{r_{-k}} \sum_{u=1}^{r_{-k}} \sum_{h \geq 0} \sum_{m \geq 0} a_{f,h} a_{f,m} X_{f,t-h,(k),ig} M_{gu} X_{f,t-m,(k),ju} \right) \\ & = \frac{1}{T^2} \sum_{t=1}^T \sum_{s=1}^T \left(\sum_{w \geq 0} a_{f,w} a_{f,w-(t-s)} \right) \left(\sum_{q \geq 0} a_{f,q} a_{f,q-(t-s)} \right) \left(\sum_{l=1}^{r_{-k}} \sum_{v=1}^{r_{-k}} M_{lv}^2 \right) \end{aligned}$$

$$\begin{aligned}
& + \frac{1}{T^2} \sum_{t=1}^T \sum_{w \geq 0} \sum_{q \geq 0} \sum_{s=1}^T \sum_{h \geq 0} \sum_{m \geq 0} a_{f,w} a_{f,q} a_{f,h} a_{f,m} \left(\sum_{l=1}^{r-k} M_{ll}^2 \right) \\
& \quad \cdot \text{Cov} \left(X_{f,t-w,(k),il} X_{f,t-q,(k),il} X_{f,s-h,(k),il} X_{f,s-m,(k),il} \right) \\
& = O\left(\frac{1}{T}\right) \cdot \left(\sum_{l=1}^{r-k} \sum_{v=1}^{r-k} M_{lv}^2 \right) + O\left(\frac{1}{T}\right) \cdot \left(\sum_{l=1}^{r-k} M_{ll}^2 \right) = O\left(\frac{1}{T}\right) \cdot \|\mathbf{M}\|_F^2 = o(1),
\end{aligned}$$

where the third equality considered $i = j$ and $i \neq j$ separately, and the fourth used (S2.4) and (S2.5). This completes the proof of (S2.1) and hence Lemma 2. $\square$

Lemma 3 (Bounding $\sum_{t=1}^T \mathbf{R}_{k,t}$) Under Assumptions (F1), (L1), (L2), (E1) and (E2), it holds that

$$\left\| \sum_{t=1}^T \mathbf{Q}_k \mathbf{F}_{Z,t,(k)} \mathbf{Q}'_{-k} \mathbf{E}'_{t,(k)} \right\|_F^2 = O_P \left(T d_k^{1+\delta_{k,1}} d_{-k} \right), \quad (\text{S2.6})$$

$$\left\| \sum_{t=1}^T \mathbf{E}_{t,(k)} \mathbf{E}'_{t,(k)} \right\|_F^2 = O_P \left(T d_k^2 d_{-k} + T^2 d_k d_{-k}^2 \right). \quad (\text{S2.7})$$

Hence, with $\mathbf{R}_{k,t}$ defined in (S2.11), we have

$$\left\| \sum_{t=1}^T \mathbf{R}_{k,t} \right\|_F^2 = O_P \left(T d_k^2 d_{-k} + T^2 d_k d_{-k}^2 \right).$$

Proof of Lemma 3. It is not hard to see (S2.6) holds as follows,

$$\begin{aligned}
& \left\| \sum_{t=1}^T \mathbf{Q}_k \mathbf{F}_{Z,t,(k)} \mathbf{Q}'_{-k} \mathbf{E}'_{t,(k)} \right\|_F^2 = \sum_{i=1}^{d_k} \sum_{l=1}^{d_k} \left(\sum_{t=1}^T \mathbf{A}'_{k,i} \mathbf{F}_{t,(k)} \mathbf{A}'_{-k} \mathbf{E}_{t,(k),l} \right)^2 \\
& = \sum_{i=1}^{d_k} \|\mathbf{A}_{k,i}\|^2 \cdot \sum_{l=1}^{d_k} \left(\sum_{h=1}^{d_k} \sum_{t=1}^T E_{t,(k),lh} \frac{1}{\|\mathbf{A}_{k,i}\|} \mathbf{A}'_{k,i} \mathbf{F}_{t,(k)} \mathbf{A}_{-k,h} \right)^2 = O_P \left(T d_k^{1+\delta_{k,1}} d_{-k} \right),
\end{aligned}$$

where the last equality is from Assumptions (L1), (L2) and Lemma 1.

Consider now (S2.7). First note from Assumption (E1) that, for any $k \in [K]$, $i \in [d_k]$, $j \in [d_{-k}]$,

$$E_{t,(k),ij} = \mathbf{A}'_{e,k,i} \mathbf{F}_{e,t,(k)} \mathbf{A}_{e,-k,j} + \Sigma_{\epsilon,(k),ij} \epsilon_{t,(k),ij},$$

where $\mathbf{A}_{e,-k} := \mathbf{A}_{e,K} \otimes \cdots \otimes \mathbf{A}_{e,k+1} \otimes \mathbf{A}_{e,k-1} \otimes \cdots \otimes \mathbf{A}_{e,1}$. Then with Assumption (E2), we have

$$\text{Cov}(E_{t,(k),ij}, E_{t,(k),lj}) = \mathbf{A}'_{e,k,i} \mathbf{A}_{e,k,l} \|\mathbf{A}_{e,-k,j}\|^2 + \Sigma_{\epsilon,(k),ij}^2 \mathbb{1}_{\{i=l\}},$$

and together with Lemma 1,

$$\begin{aligned}
\mathbb{E}\left\{\left\|\sum_{t=1}^T \mathbf{E}_{t,(k)} \mathbf{E}'_{t,(k)}\right\|_F^2\right\} &= \sum_{i=1}^{d_k} \sum_{l=1}^{d_k} \mathbb{E}\left\{\left(\sum_{t=1}^T \sum_{j=1}^{d_k} E_{t,(k),ij} E_{t,(k),lj}\right)^2\right\} \\
&= \sum_{i=1}^{d_k} \sum_{l=1}^{d_k} \left\{ \sum_{t=1}^T \sum_{j=1}^{d_k} \sum_{s=1}^T \sum_{h=1}^{d_k} \text{Cov}(E_{t,(k),ij} E_{t,(k),lj}, E_{s,(k),ih} E_{s,(k),lh}) + \left(\sum_{t=1}^T \sum_{j=1}^{d_k} \mathbb{E}[E_{t,(k),ij} E_{t,(k),lj}]\right)^2 \right\} \\
&= O(T d_k^2 d_k) + \sum_{i=1}^{d_k} \sum_{l=1}^{d_k} O\left(T \cdot \mathbf{A}'_{e,k,i} \mathbf{A}_{e,k,l} \cdot \|\mathbf{A}_{e,-k}\|_F^2 + T d_k \cdot \mathbb{1}_{\{i=l\}}\right)^2 = O(T d_k^2 d_k + T^2 d_k d_k^2).
\end{aligned}$$

Recall from (S2.11) it is defined that

$$\mathbf{R}_{k,t} = \mathbf{Q}_k \mathbf{F}_{Z,t,(k)} \mathbf{Q}'_k \mathbf{E}'_{t,(k)} + \mathbf{E}_{t,(k)} \mathbf{Q}_{-k} \mathbf{F}'_{Z,t,(k)} \mathbf{Q}'_k + \mathbf{E}_{t,(k)} \mathbf{E}'_{t,(k)},$$

the rate on $\|\sum_{t=1}^T \mathbf{R}_{k,t}\|_F^2$ is direct from (S2.6) and (S2.7). Hence the proof of Lemma 3 is complete. $\square$

Lemma 4 *Let Assumptions (F1), (L1), (L2), (E1), (E2) and (R1) hold. For $k \in [K]$, define $\widehat{\mathbf{D}}_k$ as the $r_k \times r_k$ diagonal matrix with the first largest r_k eigenvalues of $T^{-1} \sum_{t=1}^T \mathbf{Y}_{t,(k)} \mathbf{Y}'_{t,(k)}$ on the main diagonal, such that $\widehat{\mathbf{D}}_k = \widehat{\mathbf{Q}}'_k (T^{-1} \sum_{t=1}^T \mathbf{Y}_{t,(k)} \mathbf{Y}'_{t,(k)}) \widehat{\mathbf{Q}}_k$. Define $\omega_k := d_k^{\delta_k, r_k} g_s^{-\delta_k, 1}$. It holds that*

$$\|\widehat{\mathbf{D}}_k^{-1}\|_F = O_P(\omega_k^{-1}).$$

Proof of Lemma 4. Observe that $\widehat{\mathbf{D}}_k$ has size $r_k \times r_k$, it suffices to find the lower bound of $\lambda_{r_k}(\widehat{\mathbf{D}}_k)$.

To do this, consider the decomposition

$$\frac{1}{T} \sum_{t=1}^T \mathbf{Y}_{t,(k)} \mathbf{Y}'_{t,(k)} = \frac{1}{T} \sum_{t=1}^T \mathbf{Q}_k \mathbf{F}_{Z,t,(k)} \mathbf{Q}'_k \mathbf{Q}_{-k} \mathbf{F}'_{Z,t,(k)} \mathbf{Q}'_k + \frac{1}{T} \sum_{t=1}^T \mathbf{R}_{k,t}, \quad (\text{S2.8})$$

which is direct from (S2.11). Then for a unit vector $\gamma \in \mathbb{R}^{d_k}$, we can define

$$\begin{aligned}
S_k(\gamma) &:= \frac{1}{\omega_k} \gamma' \left(\frac{1}{T} \sum_{t=1}^T \mathbf{Y}_{t,(k)} \mathbf{Y}'_{t,(k)} \right) \gamma =: S_k^*(\gamma) + \tilde{S}_k(\gamma), \quad \text{with} \\
S_k^*(\gamma) &:= \frac{1}{\omega_k} \gamma' \left(\frac{1}{T} \sum_{t=1}^T \mathbf{Q}_k \mathbf{F}_{Z,t,(k)} \mathbf{Q}'_k \mathbf{Q}_{-k} \mathbf{F}'_{Z,t,(k)} \mathbf{Q}'_k \right) \gamma, \quad \tilde{S}_k(\gamma) := \frac{1}{\omega_k} \gamma' \left(\frac{1}{T} \sum_{t=1}^T \mathbf{R}_{k,t} \right) \gamma.
\end{aligned}$$

Since $\|\gamma\| = 1$, we have by Lemma 3,

$$|\tilde{S}_k(\gamma)|^2 \leq \frac{1}{\omega_k^2 T^2} \left\| \sum_{t=1}^T \mathbf{R}_{k,t} \right\|_F^2 = O_P\left(T^{-1} d_k^{1+2(\delta_k, 1 - \delta_k, r_k)} d g_s^{-2} + d_k^{2(\delta_k, 1 - \delta_k, r_k - 1/2)} d^2 g_s^{-2}\right) = O_P(1),$$

where the last equality used Assumption (R1). Next, with Assumption (F1) and Lemma 2, consider

$$\begin{aligned} \lambda_{r_k} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{Q}_k \mathbf{F}_{Z,t,(k)} \mathbf{Q}'_{-k} \mathbf{Q}_{-k} \mathbf{F}'_{Z,t,(k)} \mathbf{Q}'_k \right) &= \lambda_{r_k} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{A}_k \mathbf{F}_{t,(k)} \mathbf{A}'_{-k} \mathbf{A}_{-k} \mathbf{F}'_{t,(k)} \mathbf{A}'_k \right) \\ &\geq \lambda_{r_k} (\mathbf{A}'_k \mathbf{A}_k) \cdot \lambda_{r_k} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{F}_{t,(k)} \mathbf{A}'_{-k} \mathbf{A}_{-k} \mathbf{F}'_{t,(k)} \right) \asymp_P d_k^{\delta_{k,r_k}} \cdot \lambda_{r_k} (\text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}) \mathbf{I}_{r_k}) \asymp_P \omega_k. \end{aligned} \quad (\text{S2.9})$$

With this, going back to the decomposition (S2.8),

$$\begin{aligned} \omega_k^{-1} \lambda_{r_k} (\widehat{\mathbf{D}}_k) &= \omega_k^{-1} \lambda_{r_k} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{Y}_{t,(k)} \mathbf{Y}'_{t,(k)} \right) \\ &\geq \omega_k^{-1} \lambda_{r_k} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{Q}_k \mathbf{F}_{Z,t,(k)} \mathbf{Q}'_{-k} \mathbf{Q}_{-k} \mathbf{F}'_{Z,t,(k)} \mathbf{Q}'_k \right) - \sup_{\|\gamma\|=1} |\widetilde{S}_k(\gamma)| \asymp_P 1, \end{aligned}$$

which implies $\|\widehat{\mathbf{D}}_k^{-1}\|_F = O_P(\lambda_{r_k}^{-1}(\widehat{\mathbf{D}}_k)) = O_P(\omega_k^{-1})$, which completes the proof of Lemma 4. $\square$

Lemma 5 (*Consistency of $\{\widehat{\mathbf{Q}}_k\}_{k \in [K]}$*) *Let Assumptions (F1), (L1), (L2), (E1), (E2) and (R1) hold. For any $k \in [K]$, define an $r_k \times r_k$ matrix*

$$\mathbf{H}_k := T^{-1} \widehat{\mathbf{D}}_k^{-1} \widehat{\mathbf{Q}}'_k \mathbf{Q}_k \sum_{t=1}^T (\mathbf{F}_{Z,t,(k)} \mathbf{Q}'_{-k} \mathbf{Q}_{-k} \mathbf{F}'_{Z,t,(k)}).$$

As $T, d_1, \dots, d_K \rightarrow \infty$ we have $\mathbf{H}_k$ invertible with $\|\mathbf{H}_k\|_F = O_P(1)$ and

$$\|\widehat{\mathbf{Q}}_k - \mathbf{Q}_k \mathbf{H}'_k\|_F^2 = O_P \left\{ d_k^{2(\delta_{k,1} - \delta_{k,r_k})} \left(\frac{1}{T d_{-k} d_k^{1 - \delta_{k,1}}} + \frac{1}{d_k^{1 + \delta_{k,r_k}}} \right) \frac{d^2}{g_s^2} \right\}.$$

Proof of Lemma 5. First, we may write (3.5) as

$$\mathcal{Y}_t = \mathcal{F}_{Z,t} \times_1 \mathbf{Q}_1 \times_2 \cdots \times_K \mathbf{Q}_K + \mathcal{E}_t. \quad (\text{S2.10})$$

For any $k \in [K]$, taking the mode- k unfolding on (S2.10), we have

$$\mathbf{Y}_{t,(k)} = \mathbf{Q}_k \mathbf{F}_{Z,t,(k)} \mathbf{Q}'_{-k} + \mathbf{E}_{t,(k)},$$

where $\mathbf{F}_{Z,t,(k)}$ denotes the mode- k unfolding of $\mathcal{F}_{Z,t}$. Hence,

$$\begin{aligned} \mathbf{Y}_{t,(k)} \mathbf{Y}'_{t,(k)} &= \mathbf{Q}_k \mathbf{F}_{Z,t,(k)} \mathbf{Q}'_{-k} \mathbf{Q}_{-k} \mathbf{F}'_{Z,t,(k)} \mathbf{Q}'_k + \mathbf{Q}_k \mathbf{F}_{Z,t,(k)} \mathbf{Q}'_{-k} \mathbf{E}'_{t,(k)} + \mathbf{E}_{t,(k)} \mathbf{Q}_{-k} \mathbf{F}'_{Z,t,(k)} \mathbf{Q}'_k + \mathbf{E}_{t,(k)} \mathbf{E}'_{t,(k)} \\ &=: \mathbf{Q}_k \mathbf{F}_{Z,t,(k)} \mathbf{Q}'_{-k} \mathbf{Q}_{-k} \mathbf{F}'_{Z,t,(k)} \mathbf{Q}'_k + \mathbf{R}_{k,t}. \end{aligned} \quad (\text{S2.11})$$

Recall from Lemma 4 that $\widehat{\mathbf{D}}_k$ is the $r_k \times r_k$ diagonal matrix with the first largest r_k eigenvalues of $T^{-1} \sum_{t=1}^T \mathbf{Y}_{t,(k)} \mathbf{Y}'_{t,(k)}$ on the main diagonal, and since $\widehat{\mathbf{Q}}_k$ consists of the corresponding eigenvectors, we have

$$\widehat{\mathbf{Q}}_k \widehat{\mathbf{D}}_k = \frac{1}{T} \sum_{t=1}^T \mathbf{Y}_{t,(k)} \mathbf{Y}'_{t,(k)} \widehat{\mathbf{Q}}_k. \quad (\text{S2.12})$$

With (S2.11), we can write the j -th row of estimated mode- k factor loading as

$$\begin{aligned} \widehat{\mathbf{Q}}_{k,j\cdot} &= \frac{1}{T} \widehat{\mathbf{D}}_k^{-1} \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{k,i\cdot} \sum_{t=1}^T (\mathbf{Y}_{t,(k)} \mathbf{Y}'_{t,(k)})_{ij} \\ &= \frac{1}{T} \widehat{\mathbf{D}}_k^{-1} \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{k,i\cdot} \mathbf{Q}'_{k,i\cdot} \sum_{t=1}^T (\mathbf{F}_{Z,t,(k)} \mathbf{Q}'_{-k} \mathbf{Q}_{-k} \mathbf{F}'_{Z,t,(k)}) \mathbf{Q}_{k,j\cdot} + \frac{1}{T} \widehat{\mathbf{D}}_k^{-1} \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{k,i\cdot} \sum_{t=1}^T (\mathbf{R}_{k,t})_{ij}. \end{aligned}$$

Hence with the definition $\mathbf{H}_k = T^{-1} \widehat{\mathbf{D}}_k^{-1} \widehat{\mathbf{Q}}'_k \mathbf{Q}_k \sum_{t=1}^T (\mathbf{F}_{Z,t,(k)} \mathbf{Q}'_{-k} \mathbf{Q}_{-k} \mathbf{F}'_{Z,t,(k)})$, we may decompose

$$\widehat{\mathbf{Q}}_{k,j\cdot} - \mathbf{H}_k \mathbf{Q}_{k,j\cdot} = \frac{1}{T} \widehat{\mathbf{D}}_k^{-1} \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{k,i\cdot} \sum_{t=1}^T (\mathbf{R}_{k,t})_{ij} \quad (\text{S2.13})$$

$$= \frac{1}{T} \widehat{\mathbf{D}}_k^{-1} \sum_{i=1}^{d_k} (\widehat{\mathbf{Q}}_{k,i\cdot} - \mathbf{H}_k \mathbf{Q}_{k,i\cdot}) \sum_{t=1}^T (\mathbf{R}_{k,t})_{ij} + \frac{1}{T} \widehat{\mathbf{D}}_k^{-1} \sum_{i=1}^{d_k} \mathbf{H}_k \mathbf{Q}_{k,i\cdot} \sum_{t=1}^T (\mathbf{R}_{k,t})_{ij}. \quad (\text{S2.14})$$

With the decomposition (S2.13), it holds that

$$\begin{aligned} \|\widehat{\mathbf{Q}}_k - \mathbf{Q}_k \mathbf{H}'_k\|_F^2 &= \sum_{j=1}^{d_k} \|\widehat{\mathbf{Q}}_{k,j\cdot} - \mathbf{H}_k \mathbf{Q}_{k,j\cdot}\|^2 = \sum_{j=1}^{d_k} \left\| \frac{1}{T} \widehat{\mathbf{D}}_k^{-1} \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{k,i\cdot} \sum_{t=1}^T (\mathbf{R}_{k,t})_{ij} \right\|^2 \\ &= \sum_{j=1}^{d_k} \left\| \frac{1}{T} \widehat{\mathbf{D}}_k^{-1} \widehat{\mathbf{Q}}'_k \left(\sum_{t=1}^T \mathbf{R}_{k,t} \right)_j \right\|^2 \leq \frac{1}{T^2} \cdot \|\widehat{\mathbf{D}}_k^{-1}\|_F^2 \cdot \|\widehat{\mathbf{Q}}_k\|_F^2 \cdot \left\| \sum_{t=1}^T \mathbf{R}_{k,t} \right\|_F^2 \\ &= O_P \left(d_k^{2(\delta_{k,1} - \delta_{k,r_k})} \left(\frac{1}{T d_k} + \frac{1}{d_k} \right) \frac{d^2}{g_s^2} \right) = o_P(1), \end{aligned} \quad (\text{S2.15})$$

where the second last equality used Lemmas 3 and 4, and the last used Assumption (R1).

Before improving the rate of (S2.15), we now use it to show $\mathbf{H}_k$ has full rank and $\|\mathbf{H}_k\|_F = O_P(1)$ asymptotically. To this end, it is sufficient to observe

$$\begin{aligned} \mathbf{I}_{r_k} &= \widehat{\mathbf{Q}}'_k \widehat{\mathbf{Q}}_k = \widehat{\mathbf{Q}}'_k (\widehat{\mathbf{Q}}_k - \mathbf{Q}_k \mathbf{H}'_k) + \widehat{\mathbf{Q}}'_k \mathbf{Q}_k \mathbf{H}'_k = \widehat{\mathbf{Q}}'_k \widehat{\mathbf{Q}}_k \mathbf{H}'_k + o_P(1) \\ &= \mathbf{H}_k \mathbf{Q}'_k \mathbf{Q}_k \mathbf{H}'_k + o_P(1) = \mathbf{H}_k \boldsymbol{\Sigma}_{A,k} \mathbf{H}'_k + o_P(1), \end{aligned}$$

where the last equality used Assumption (L1) or (L2) and it is immediate that $\mathbf{H}_k$ has full rank asymp-

totically. Let $\sigma_i(\mathbf{X})$ denote the i -th largest singular value for any give matrix $\mathbf{X}$, we have

$$\sigma_1(\mathbf{H}_k) \cdot \sigma_{r_k}(\mathbf{\Sigma}_{A,k}) \cdot \sigma_{r_k}(\mathbf{H}_k^T) \leq \sigma_1(\mathbf{H}_k) \cdot \sigma_{r_k}(\mathbf{\Sigma}_{A,k} \mathbf{H}_k^T) \leq \sigma_1(\mathbf{H}_k \mathbf{\Sigma}_{A,k} \mathbf{H}_k') = O_P(\sigma_1(\mathbf{I}_{r_k})) = O_P(1),$$

which implies $\|\mathbf{H}_k\|_F = O_P(1)$ by Assumption (L1) or (L2).

Consider the decomposition (S2.14), we have

$$\begin{aligned} \|\widehat{\mathbf{Q}}_k - \mathbf{Q}_k \mathbf{H}_k'\|_F^2 &= \sum_{j=1}^{d_k} \|\widehat{\mathbf{Q}}_{k,j\cdot} - \mathbf{H}_k \mathbf{Q}_{k,j\cdot}\|^2 \\ &= O_P \left(\sum_{j=1}^{d_k} \left\| \frac{1}{T} \widehat{\mathbf{D}}_k^{-1} \sum_{i=1}^{d_k} \mathbf{H}_k \mathbf{Q}_{k,i\cdot} \sum_{t=1}^T (\mathbf{R}_{k,t})_{ij} \right\|^2 + \sum_{j=1}^{d_k} \left\| \frac{1}{T} \widehat{\mathbf{D}}_k^{-1} \sum_{i=1}^{d_k} (\widehat{\mathbf{Q}}_{k,i\cdot} - \mathbf{H}_k \mathbf{Q}_{k,i\cdot}) \sum_{t=1}^T (\mathbf{R}_{k,t})_{ij} \right\|^2 \right) \\ &= O_P \left(\sum_{j=1}^{d_k} \left\| \frac{1}{T} \widehat{\mathbf{D}}_k^{-1} \mathbf{H}_k \sum_{i=1}^{d_k} \mathbf{Q}_{k,i\cdot} \sum_{t=1}^T (\mathbf{R}_{k,t})_{ij} \right\|^2 \right) + O_P \left(\frac{1}{T^2} \|\widehat{\mathbf{D}}_k^{-1}\|_F^2 \cdot \|\widehat{\mathbf{Q}}_k - \mathbf{Q}_k \mathbf{H}_k'\|_F^2 \cdot \left\| \sum_{t=1}^T \mathbf{R}_{k,t} \right\|_F^2 \right) \\ &=: O_P \left(T^{-2} d_k^{2(\delta_{k,1} - \delta_{k,r_k})} g_s^{-2} \right) \cdot \sum_{j=1}^{d_k} (\mathcal{I}_{1,j} + \mathcal{I}_{2,j} + \mathcal{I}_{3,j}) + O_P \left(\|\widehat{\mathbf{Q}}_k - \mathbf{Q}_k \mathbf{H}_k'\|_F^2 \right), \end{aligned} \tag{S2.16}$$

where the last equality used $\|\mathbf{H}_k\|_F = O_P(1)$, Lemmas 3 and 4, and the definitions

$$\begin{aligned} \mathcal{I}_{1,j} &:= \left\| \sum_{i=1}^{d_k} \mathbf{Q}_{k,i\cdot} \sum_{t=1}^T \mathbf{Q}'_{k,i\cdot} \mathbf{F}_{Z,t,(k)} \mathbf{Q}'_{-k} \mathbf{E}_{t,(k),j\cdot} \right\|^2, \quad \mathcal{I}_{2,j} := \left\| \sum_{i=1}^{d_k} \mathbf{Q}_{k,i\cdot} \sum_{t=1}^T \mathbf{E}'_{t,(k),i\cdot} \mathbf{Q}_{-k} \mathbf{F}'_{Z,t,(k)} \mathbf{Q}_{k,j\cdot} \right\|^2, \\ \mathcal{I}_{3,j} &:= \left\| \sum_{i=1}^{d_k} \mathbf{Q}_{k,i\cdot} \sum_{t=1}^T \mathbf{E}'_{t,(k),i\cdot} \mathbf{E}_{t,(k),j\cdot} \right\|^2. \end{aligned}$$

By the Cauchy-Schwarz inequality,

$$\begin{aligned} \mathcal{I}_{1,j} &= \left\| \sum_{i=1}^{d_k} \mathbf{Q}_{k,i\cdot} \sum_{t=1}^T \mathbf{A}'_{k,i\cdot} \mathbf{F}_{t,(k)} \mathbf{A}'_{-k} \mathbf{E}_{t,(k),j\cdot} \right\|^2 \leq \left(\sum_{i=1}^{d_k} \|\mathbf{Q}_{k,i\cdot}\|^2 \right) \left\{ \sum_{i=1}^{d_k} \left(\sum_{t=1}^T \mathbf{A}'_{k,i\cdot} \mathbf{F}_{t,(k)} \mathbf{A}'_{-k} \mathbf{E}_{t,(k),j\cdot} \right)^2 \right\} \\ &= \|\mathbf{Q}_k\|_F^2 \cdot \left\{ \sum_{i=1}^{d_k} \|\mathbf{A}_{k,i\cdot}\|^2 \left(\sum_{h=1}^{d_k} \sum_{t=1}^T \frac{\mathbf{A}'_{k,i\cdot}}{\|\mathbf{A}_{k,i\cdot}\|} \mathbf{F}_{t,(k)} \mathbf{A}_{-k,h\cdot} \mathbf{E}_{t,(k),jh} \right)^2 \right\} = O_P \left(T d_k^{\delta_{k,1}} d_{-k} \right), \end{aligned}$$

where the last equality used Assumptions (L1), (L2) and Lemma 1.

Consider $\mathcal{I}_{2,j}$. From Assumptions (E1) and (E2), we have for any $t \in [T], k \in [K], i \in [j_k], h \in [d_{-k}]$,

$$E_{t,(k),ih} = \sum_{w \geq 0} a_{e,w} \mathbf{A}'_{e,k,i\cdot} \mathbf{X}_{e,t-w,(k)} \mathbf{A}_{e,-k,h\cdot} + \Sigma_{\epsilon,(k),ih} \sum_{w \geq 0} a_{\epsilon,w} X_{\epsilon,t-w,(k),ih}.$$

By Assumptions (F1), (E1) and (E2), we first have

$$\begin{aligned}
& \mathbb{E} \left\{ \left(\sum_{t=1}^T \sum_{h=1}^{d_k} \left(\sum_{w \geq 0} a_{e,w} \mathbf{A}'_{e,k,i} \cdot \mathbf{X}_{e,t-w,(k)} \mathbf{A}_{e,-k,h} \right) \mathbf{A}'_{-k,h} \cdot \mathbf{F}'_{t,(k)} \mathbf{A}_{k,j} \right)^2 \right\} \\
&= \text{Cov} \left(\sum_{t=1}^T \sum_{h=1}^{d_k} \mathbf{A}'_{-k,h} \cdot \left(\sum_{w \geq 0} a_{f,w} \mathbf{X}'_{f,t-w,(k)} \right) \mathbf{A}_{k,j}, \left(\sum_{w \geq 0} a_{e,w} \mathbf{A}'_{e,k,i} \cdot \mathbf{X}_{e,t-w,(k)} \mathbf{A}_{e,-k,h} \right), \right. \\
&\quad \left. \sum_{t=1}^T \sum_{h=1}^{d_k} \mathbf{A}'_{-k,h} \cdot \left(\sum_{w \geq 0} a_{f,w} \mathbf{X}'_{f,t-w,(k)} \right) \mathbf{A}_{k,j}, \left(\sum_{w \geq 0} a_{e,w} \mathbf{A}'_{e,k,i} \cdot \mathbf{X}_{e,t-w,(k)} \mathbf{A}_{e,-k,h} \right) \right) \quad (\text{S2.17}) \\
&= \sum_{h=1}^{d_k} \sum_{l=1}^{d_k} \sum_{t=1}^T \sum_{w \geq 0} a_{f,w}^2 a_{e,w}^2 \cdot \|\mathbf{A}_{k,j}\|^2 \cdot \|\mathbf{A}_{-k,h}\| \cdot \|\mathbf{A}_{-k,l}\| \cdot \|\mathbf{A}_{e,-k,h}\| \cdot \|\mathbf{A}_{e,-k,l}\| \cdot \|\mathbf{A}_{e,k,i}\|^2 \\
&= O(T) \cdot \|\mathbf{A}_{k,j}\|^2 \cdot \|\mathbf{A}_{e,k,i}\|^2.
\end{aligned}$$

Similarly, it holds that

$$\begin{aligned}
& \mathbb{E} \left\{ \left\| \sum_{i=1}^{d_k} \sum_{h=1}^{d_k} \sum_{t=1}^T \mathbf{Q}_{k,i} \cdot \left(\Sigma_{\epsilon,(k),ih} \sum_{w \geq 0} a_{\epsilon,w} X_{\epsilon,t-w,(k),ih} \right) \mathbf{A}'_{-k,h} \cdot \mathbf{F}'_{t,(k)} \mathbf{A}_{k,j} \right\|^2 \right\} \\
&= \text{Cov} \left(\sum_{i=1}^{d_k} \sum_{h=1}^{d_k} \sum_{t=1}^T \mathbf{Q}_{k,i} \cdot \left(\Sigma_{\epsilon,(k),ih} \sum_{w \geq 0} a_{\epsilon,w} X_{\epsilon,t-w,(k),ih} \right) \mathbf{A}'_{-k,h} \cdot \mathbf{F}'_{t,(k)} \mathbf{A}_{k,j}, \right. \\
&\quad \left. \sum_{i=1}^{d_k} \sum_{h=1}^{d_k} \sum_{t=1}^T \mathbf{Q}_{k,i} \cdot \left(\Sigma_{\epsilon,(k),ih} \sum_{w \geq 0} a_{\epsilon,w} X_{\epsilon,t-w,(k),ih} \right) \mathbf{A}'_{-k,h} \cdot \mathbf{F}'_{t,(k)} \mathbf{A}_{k,j} \right) \quad (\text{S2.18}) \\
&= \sum_{i=1}^{d_k} \sum_{h=1}^{d_k} \sum_{t=1}^T \sum_{w \geq 0} a_{f,w}^2 a_{\epsilon,w}^2 \cdot \|\mathbf{A}_{k,j}\|^2 \cdot \|\mathbf{A}_{-k,h}\|^2 \cdot \Sigma_{\epsilon,(k),ih}^2 \cdot \|\mathbf{Q}_{k,i}\|^2 = O(T) \cdot \|\mathbf{A}_{k,j}\|^2 \cdot \|\mathbf{A}_{-k}\|^2.
\end{aligned}$$

Hence using Lemma 4, it holds that

$$\begin{aligned}
\mathcal{I}_{2,j} &= \left\| \sum_{i=1}^{d_k} \mathbf{Q}_{k,i} \cdot \sum_{t=1}^T \mathbf{E}'_{t,(k),i} \cdot \mathbf{A}_{-k} \mathbf{F}'_{t,(k)} \mathbf{A}_{k,j} \right\|^2 = \left\| \sum_{i=1}^{d_k} \mathbf{Q}_{k,i} \cdot \sum_{h=1}^{d_k} \sum_{t=1}^T E_{t,(k),ih} \mathbf{A}'_{-k,h} \cdot \mathbf{F}'_{t,(k)} \mathbf{A}_{k,j} \right\|^2 \\
&\leq 2 \left\{ \left\| \sum_{i=1}^{d_k} \sum_{h=1}^{d_k} \sum_{t=1}^T \mathbf{Q}_{k,i} \cdot \left(\Sigma_{\epsilon,(k),ih} \sum_{w \geq 0} a_{\epsilon,w} X_{\epsilon,t-w,(k),ih} \right) \mathbf{A}'_{-k,h} \cdot \mathbf{F}'_{t,(k)} \mathbf{A}_{k,j} \right\|^2 \right\} \\
&\quad + 2 \left\{ \|\mathbf{Q}_k\|^2 \cdot \sum_{i=1}^{d_k} \left(\sum_{t=1}^T \sum_{h=1}^{d_k} \left(\sum_{w \geq 0} a_{e,w} \mathbf{A}'_{e,k,i} \cdot \mathbf{X}_{e,t-w,(k)} \mathbf{A}_{e,-k,h} \right) \mathbf{A}'_{-k,h} \cdot \mathbf{F}'_{t,(k)} \mathbf{A}_{k,j} \right)^2 \right\} \\
&= O_P \left(\|\mathbf{A}_{k,j}\|^2 \cdot T d_k^{-\delta_{k,1}} g_s \right)
\end{aligned}$$

where we used (S2.17) and (S2.18) in the last equality.

For $\mathcal{I}_{3,j}$, let $r_{e,-k} := \prod_{p \neq k} r_{e,p}$. Then from the noise structure in Assumptions (E1) and (E2),

$$\begin{aligned}
& \text{Var} \left(\sum_{i=1}^{d_k} \sum_{h=1}^{d_{-k}} \sum_{t=1}^T \mathbf{Q}_{k,i} \cdot E_{t,(k),ih} E_{t,(k),jh} \right) \\
&= O \left(\sum_{i=1}^{d_k} \sum_{u=1}^{d_k} \sum_{h=1}^{d_{-k}} \sum_{l=1}^{d_{-k}} \sum_{t=1}^T \sum_{n=1}^{r_{e,k}} \sum_{m=1}^{r_{e,-k}} \sum_{w \geq 0} d_{e,w}^4 A_{e,k,in} A_{e,k,un} A_{e,k,jn}^2 A_{e,-k,hm}^2 A_{e,-k,lm}^2 \right. \\
&\quad \cdot \|\mathbf{Q}_{k,i}\| \cdot \|\mathbf{Q}_{k,u}\| \cdot \text{Var}(X_{e,t-w,(k),nm}^2) \Big) \\
&\quad + O \left(\sum_{i=1}^{d_k} \sum_{h=1}^{d_{-k}} \sum_{t=1}^T \sum_{w \geq 0} a_{e,w}^4 \Sigma_{e,(k),ih}^2 \Sigma_{e,(k),jh}^2 \cdot \|\mathbf{Q}_{k,i}\|^2 \cdot \text{Var}(X_{e,t-w,(k),ih} X_{e,t-w,(k),jh}) \right) \\
&= O(T + T d_{-k}) = O(T d_{-k}).
\end{aligned}$$

Moreover, it also holds that

$$\begin{aligned}
& \mathbb{E} \left(\sum_{i=1}^{d_k} \left| \sum_{h=1}^{d_{-k}} \sum_{t=1}^T E_{t,(k),ih} E_{t,(k),jh} \right| \right) \\
&= \sum_{i=1}^{d_k} \left| \sum_{h=1}^{d_{-k}} \sum_{t=1}^T \left(\|\mathbf{A}_{e,-k,h}\|^2 \cdot \|\mathbf{A}_{e,k,i}\| \cdot \|\mathbf{A}_{e,k,j}\| + \Sigma_{e,(k),ih} \mathbb{1}_{\{i=j\}} \right) \right| = O(T d_{-k}),
\end{aligned}$$

together with $\max_i \|\mathbf{Q}_{k,i}\|^2 \leq \|\mathbf{A}_{k,j}\|^2 \cdot \|\mathbf{Z}_k^{-1/2}\|^2 = O_P(d_k^{-\delta_k, r_k})$, we arrive at

$$\mathcal{I}_{3,j} = \left\| \sum_{i=1}^{d_k} \sum_{h=1}^{d_{-k}} \mathbf{Q}_{k,i} \cdot \sum_{t=1}^T E_{t,(k),ih} E_{t,(k),jh} \right\|^2 = O_P(T d_{-k} + T^2 d_k^{-\delta_k, r_k} d_{-k}^2).$$

Finally, for (S2.16) we have

$$\begin{aligned}
\|\hat{\mathbf{Q}}_k - \mathbf{Q}_k \mathbf{H}_k'\|_F^2 &= O_P \left\{ T^{-2} d_k^{2(\delta_{k,1} - \delta_k, r_k)} g_s^{-2} \cdot \left(T d d_k^{\delta_{k,1}} + T^2 d_k^{1 - \delta_k, r_k} d_{-k}^2 \right) \right\} + o_P \left(\|\hat{\mathbf{Q}}_k - \mathbf{Q}_k \mathbf{H}_k'\|_F^2 \right) \\
&= O_P \left\{ d_k^{2(\delta_{k,1} - \delta_k, r_k)} \left(\frac{1}{T d_{-k} d_k^{1 - \delta_{k,1}}} + \frac{1}{d_k^{1 + \delta_k, r_k}} \right) \frac{d^2}{g_s^2} \right\}.
\end{aligned}$$

This completes the proof of Lemma 5. $\square$

Lemma 6 (*Consistency of $\hat{\mathcal{C}}_t$*) *Let Assumptions in Lemma 5 hold. With $\{\mathbf{Z}_k\}_{k \in [K]}$ from Assumptions (L1) and (L2) and $\{\mathbf{H}_k\}_{k \in [K]}$ from the statement of Lemma 5, define*

$$\mathbf{Z}_{\otimes} := \mathbf{Z}_K \otimes \cdots \otimes \mathbf{Z}_1, \quad \mathbf{H}_{\otimes} := \mathbf{H}_K \otimes \cdots \otimes \mathbf{H}_1, \quad \text{vec}(\hat{\mathcal{F}}_t) := (\hat{\mathbf{Q}}_K \otimes \cdots \otimes \hat{\mathbf{Q}}_1)' \text{vec}(\mathcal{Y}_t).$$

Then the estimators of the vectorised (renormalised) core factor and the i -th entry of the vectorised common

component for (3.5) are consistent such that

$$\begin{aligned} & \|\mathbf{vec}(\widehat{\mathcal{F}}_t) - (\mathbf{H}'_{\otimes})^{-1} \mathbf{vec}(\mathcal{F}_{Z,t})\|^2 = \|\mathbf{vec}(\widehat{\mathcal{F}}_t) - (\mathbf{H}'_{\otimes})^{-1} \mathbf{Z}_{\otimes}^{1/2} \mathbf{vec}(\mathcal{F}_t)\|^2 \\ & = O_P \left(\max_{k \in [K]} \left\{ d_k^{2(\delta_{k,1} - \delta_{k,r_k})} \left(\frac{1}{T d_{-k} d_k^{1-\delta_{k,1}}} + \frac{1}{d_k^{1+\delta_{k,r_k}}} \right) \frac{d^2}{g_s} \right\} + \frac{d}{g_w} \right), \end{aligned} \quad (\text{S2.19})$$

$$\begin{aligned} & \left\{ (\mathbf{vec}(\widehat{\mathcal{C}}_t))_i - (\mathbf{vec}(\mathcal{C}_t))_i \right\}^2 \\ & = O_P \left(\max_{k \in [K]} \left\{ d_k^{2(\delta_{k,1} - \delta_{k,r_k})} \left(\frac{1}{T d_{-k} d_k^{1-\delta_{k,1}}} + \frac{1}{d_k^{1+\delta_{k,r_k}}} \right) \frac{d^2}{g_s g_w} \right\} + \frac{d}{g_w^2} \right). \end{aligned} \quad (\text{S2.20})$$

Proof of Lemma 6. By (3.5), we have

$$\begin{aligned} & \mathbf{vec}(\widehat{\mathcal{F}}_t) - (\mathbf{H}'_{\otimes})^{-1} \mathbf{Z}_{\otimes}^{1/2} \mathbf{vec}(\mathcal{F}_t) = (\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1)' \mathbf{vec}(\mathcal{Y}_t) - (\mathbf{H}'_{\otimes})^{-1} \mathbf{Z}_{\otimes}^{1/2} \mathbf{vec}(\mathcal{F}_t) \\ & = (\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1)' \mathbf{vec}(\mathcal{F}_t \times_{k=1}^K \mathbf{A}_k + \mathcal{E}_t) - (\mathbf{H}'_{\otimes})^{-1} \mathbf{Z}_{\otimes}^{1/2} \mathbf{vec}(\mathcal{F}_t) \\ & = (\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1)' \left\{ (\mathbf{Q}_K \mathbf{H}'_K \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}'_1) - (\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) \right\} (\mathbf{H}'_{\otimes})^{-1} \mathbf{Z}_{\otimes}^{1/2} \mathbf{vec}(\mathcal{F}_t) \\ & \quad + \left\{ (\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) - (\mathbf{Q}_K \mathbf{H}'_K \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}'_1) \right\}' \mathbf{vec}(\mathcal{E}_t) \\ & \quad + (\mathbf{Q}_K \mathbf{H}'_K \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}'_1)' \mathbf{vec}(\mathcal{E}_t) =: \mathcal{I}_{f,1} + \mathcal{I}_{f,2} + \mathcal{I}_{f,3}. \end{aligned} \quad (\text{S2.21})$$

We first show by an induction argument that for any positive integer K ,

$$\|(\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) - (\mathbf{Q}_K \mathbf{H}'_K \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}'_1)\|_F = O_P \left(\max_{k \in [K]} \|\widehat{\mathbf{Q}}_k - \mathbf{Q}_k \mathbf{H}'_k\|_F \right). \quad (\text{S2.22})$$

The initial case for $K = 1$ is trivial. Suppose (S2.22) holds for $K - 1$, then

$$\begin{aligned} & \|(\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) - (\mathbf{Q}_K \mathbf{H}'_K \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}'_1)\|_F \\ & = \|(\widehat{\mathbf{Q}}_K - \mathbf{Q}_K \mathbf{H}'_K) \otimes (\widehat{\mathbf{Q}}_{K-1} \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) \\ & \quad + \mathbf{Q}_K \mathbf{H}'_K \otimes \{(\widehat{\mathbf{Q}}_{K-1} \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) - (\mathbf{Q}_{K-1} \mathbf{H}'_{K-1} \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}'_1)\}\|_F \\ & = O_P \left(\|\widehat{\mathbf{Q}}_K - \mathbf{Q}_K \mathbf{H}'_K\|_F + \|(\widehat{\mathbf{Q}}_{K-1} \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) - (\mathbf{Q}_{K-1} \mathbf{H}'_{K-1} \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}'_1)\|_F \right), \end{aligned}$$

which concludes (S2.22). Hence for $\mathcal{I}_{f,1}$, with Lemma 5 we immediately have

$$\begin{aligned} \|\mathcal{I}_{f,1}\|^2 & = \|(\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1)' \left\{ (\mathbf{Q}_K \mathbf{H}'_K \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}'_1) - (\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) \right\} (\mathbf{H}'_{\otimes})^{-1} \mathbf{Z}_{\otimes}^{1/2} \mathbf{vec}(\mathcal{F}_t)\|^2 \\ & = O_P \left(\|(\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) - (\mathbf{Q}_K \mathbf{H}'_K \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}'_1)\|_F^2 \cdot \|\mathbf{Z}_{\otimes}^{1/2} \mathbf{vec}(\mathcal{F}_t)\|^2 \right) \\ & = O_P \left(\max_{k \in [K]} \left\{ d_k^{2(\delta_{k,1} - \delta_{k,r_k})} \left(\frac{1}{T d_{-k} d_k^{1-\delta_{k,1}}} + \frac{1}{d_k^{1+\delta_{k,r_k}}} \right) \frac{d^2}{g_s} \right\} \right), \end{aligned} \quad (\text{S2.23})$$

where the last equality used (S2.22), Assumptions (F1), (L1) and (L2).

For $\mathcal{I}_{f,2}$, observe first throughout the proof of Lemma 5, the consistency of $\widehat{\mathbf{Q}}_{k,j\cdot}$ for $k \in [K], j \in [d_k]$ can be shown with the same argument (omitted), before eventually being aggregated over all d_k rows. That is,

$$\|\widehat{\mathbf{Q}}_{k,j\cdot} - \mathbf{H}_k \mathbf{Q}_{k,j\cdot}\|^2 = O_P \left\{ d_k^{2(\delta_{k,1} - \delta_{k,r_k}) - 1} \left(\frac{1}{T d_{-k} d_k^{1 - \delta_{k,1}}} + \frac{1}{d_k^{1 + \delta_{k,r_k}}} \right) \frac{d^2}{g_s^2} \right\}. \quad (\text{S2.24})$$

Then we have $\|\widehat{\mathbf{Q}}_{k,j\cdot}\|^2 = O_P(\|\widehat{\mathbf{Q}}_{k,j\cdot} - \mathbf{H}_k \mathbf{Q}_{k,j\cdot}\|^2 + \|\mathbf{H}_k \mathbf{Z}_k^{-1/2} \mathbf{A}_{k,j\cdot}\|^2) = O_P(d_k^{-\delta_{k,r_k}})$ which used (S2.24), Assumptions (L1) (or (L2)) and (R1). Note that this rate $d_k^{-\delta_{k,r_k}}$ is the same as the one for $\|\mathbf{Q}_{k,j\cdot}\|^2$, shown in the proof of Lemma 5. Moreover, it holds that for any positive integer k that

$$\begin{aligned} & \max_{\ell \in [d]} \left\| [(\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) - (\mathbf{Q}_K \mathbf{H}'_K \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}'_1)]_{\ell} \right\|^2 \\ &= O \left(\max_{k \in [K]} \left\{ \max_{j \in [d_k]} \|\widehat{\mathbf{Q}}_{k,j\cdot} - \mathbf{H}_k \mathbf{Q}_{k,j\cdot}\|^2 \cdot \prod_{j \in [K] \setminus \{k\}} \max_{i \in [d_j]} \|\widehat{\mathbf{Q}}_{j,i\cdot}\|^2 \right\} \right), \end{aligned} \quad (\text{S2.25})$$

which can be shown by an induction argument for which the initial case for $K = 1$ is trivial, and the induction step is seen by

$$\begin{aligned} & \max_{\ell \in [d]} \left\| [(\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) - (\mathbf{Q}_K \mathbf{H}'_K \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}'_1)]_{\ell} \right\|^2 \\ &= O \left(\max_{j \in [d_K]} \|\widehat{\mathbf{Q}}_{K,j\cdot} - \mathbf{H}_K \mathbf{Q}_{K,j\cdot}\|^2 \cdot \prod_{k=1}^{K-1} \max_{i \in [d_k]} \|\widehat{\mathbf{Q}}_{k,i\cdot}\|^2 \right. \\ & \quad \left. + \max_{i \in [d_K]} \|\mathbf{Q}_{K,i\cdot}\|^2 \max_{i \in [\prod_{k=1}^{K-1} d_k]} \left\| [(\widehat{\mathbf{Q}}_{K-1} \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) - (\mathbf{Q}_{K-1} \mathbf{H}'_{K-1} \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}'_1)]_i \right\|^2 \right). \end{aligned}$$

Hence, for $\mathcal{I}_{f,2}$ we have

$$\begin{aligned} \|\mathcal{I}_{f,2}\|^2 &= \left\| \{(\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) - (\mathbf{Q}_K \mathbf{H}'_K \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}'_1)\}' \mathbf{vec}(\mathcal{E}_t) \right\|^2 \\ &= \max_{\ell \in [d]} \left\| [(\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) - (\mathbf{Q}_K \mathbf{H}'_K \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}'_1)]_{\ell} \right\|^2 \sum_{j,l=1}^d \left| \mathbb{E}(\mathbf{vec}(\mathcal{E}_t))_j (\mathbf{vec}(\mathcal{E}_t))_l \right| \quad (\text{S2.26}) \\ &= O_P \left(\max_{k \in [K]} \left\{ d_k^{2\delta_{k,1} - \delta_{k,r_k}} \left(\frac{1}{T d_{-k} d_k^{2 - \delta_{k,1}}} + \frac{1}{d_k^{2 + \delta_{k,r_k}}} \right) \frac{d^3}{g_s^2 g_w} \right\} \right), \end{aligned}$$

where the last equality used (S2.24), (S2.25) and the first result on Lemma 1.1.

Lastly, consider $\mathcal{I}_{f,3}$. With Assumptions (L1) and (L2), we have

$$\begin{aligned}
\|\mathcal{I}_{f,3}\|^2 &= \left\| (\mathbf{Q}_K \mathbf{H}'_K \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}'_1)' \mathbf{vec}(\mathcal{E}_t) \right\|^2 = O_P \left(\left\| (\mathbf{Q}'_K \otimes \cdots \otimes \mathbf{Q}'_1) \mathbf{vec}(\mathcal{E}_t) \right\|^2 \right) \\
&= O_P \left(\left\| \mathbf{Z}_{\otimes}^{-1/2} \right\|_F^2 \cdot \left\| (\mathbf{A}'_K \otimes \cdots \otimes \mathbf{A}'_1) \mathbf{vec}(\mathcal{E}_t) \right\|^2 \right) \\
&= O_P \left\{ g_w^{-1} \cdot \sum_{j=1}^d \left\| (\mathbf{vec}(\mathcal{E}_t))_j (\mathbf{A}_K \otimes \cdots \otimes \mathbf{A}_1)_{j\cdot} \right\|^2 \right\} = O_P(d/g_w),
\end{aligned} \tag{S2.27}$$

where the last equality used (L1) and (L2) and the first result on Lemma 1.1 that

$$\begin{aligned}
&\mathbb{E} \left\{ \sum_{j=1}^d \left\| (\mathbf{vec}(\mathcal{E}_t))_j (\mathbf{A}_K \otimes \cdots \otimes \mathbf{A}_1)_{j\cdot} \right\|^2 \right\} \\
&\leq \max_{j \in [d]} \left\| (\mathbf{A}_K \otimes \cdots \otimes \mathbf{A}_1)_{j\cdot} \right\|^2 \sum_{j,l=1}^d \left| \mathbb{E} (\mathbf{vec}(\mathcal{E}_t))_j (\mathbf{vec}(\mathcal{E}_t))_l \right| = O(d).
\end{aligned}$$

Combining (S2.21), (S2.23), (S2.26) and (S2.27), we obtain (S2.19).

It remains to show (S2.20). To this end, from (3.5) and (3.9) (where $\widehat{\mathcal{C}}_{m,t}$ is simplified as $\widehat{\mathcal{C}}_t$, explained in the summary of proofs), we have

$$\begin{aligned}
\mathbf{vec}(\widehat{\mathcal{C}}_t) - \mathbf{vec}(\mathcal{C}_t) &= (\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) \mathbf{vec}(\widehat{\mathcal{F}}_t) - \mathbf{vec}(\mathcal{F}_t \times_{k=1}^K \mathbf{A}_k) \\
&= (\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) \mathbf{vec}(\widehat{\mathcal{F}}_t) - (\mathbf{Q}_K \mathbf{H}'_K \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}'_1) (\mathbf{H}'_{\otimes})^{-1} \mathbf{Z}_{\otimes}^{1/2} \mathbf{vec}(\mathcal{F}_t) \\
&= (\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) \left\{ \mathbf{vec}(\widehat{\mathcal{F}}_t) - (\mathbf{H}'_{\otimes})^{-1} \mathbf{Z}_{\otimes}^{1/2} \mathbf{vec}(\mathcal{F}_t) \right\} \\
&\quad + \left\{ (\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) - (\mathbf{Q}_K \mathbf{H}'_K \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}'_1) \right\} (\mathbf{H}'_{\otimes})^{-1} \mathbf{Z}_{\otimes}^{1/2} \mathbf{vec}(\mathcal{F}_t),
\end{aligned}$$

which implies that

$$\begin{aligned}
&\left\{ (\mathbf{vec}(\widehat{\mathcal{C}}_t))_i - (\mathbf{vec}(\mathcal{C}_t))_i \right\}^2 \\
&= O_P \left(\prod_{k=1}^K \max_{j \in [d_k]} \left\| \widehat{\mathbf{Q}}_{k,j\cdot} \right\|^2 \cdot \left\| \mathbf{vec}(\widehat{\mathcal{F}}_t) - (\mathbf{H}'_{\otimes})^{-1} \mathbf{Z}_{\otimes}^{1/2} \mathbf{vec}(\mathcal{F}_t) \right\|^2 \right. \\
&\quad \left. + \max_{\ell \in [d]} \left\| [(\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) - (\mathbf{Q}_K \mathbf{H}'_K \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}'_1)]_{\ell\cdot} \right\|^2 \cdot \left\| \mathbf{Z}_{\otimes}^{1/2} \right\|_F^2 \right) \\
&= O_P \left(\max_{k \in [K]} \left\{ d_k^{2(\delta_{k,1} - \delta_{k,r_k})} \left(\frac{1}{T d_k d_k^{1-\delta_{k,1}}} + \frac{1}{d_k^{1+\delta_{k,r_k}}} \right) \frac{d^2}{g_s g_w} \right\} + \frac{d}{g_w^2} \right),
\end{aligned}$$

where the last equality used (S2.19), (S2.24), (S2.25), Assumptions (L1) and (L2). This completes the proof of Lemma 6. $\square$

Lemma 7 (Consistency of $\{\widetilde{\mathbf{Q}}_j\}_{j \in [v-1]}$, $\widetilde{\mathbf{Q}}_V$ and $\widetilde{\mathcal{C}}_{\text{reshape},t}$) Let Assumptions (F1), (L1), (E1), (E2) and

(R1) hold, and consider the model (3.3). For $j \in [v-1]$, define $\tilde{\mathbf{D}}_j$ as the $r_j \times r_j$ diagonal matrix with the first largest r_j eigenvalues of

$$\frac{1}{T} \sum_{t=1}^T \text{Reshape}(\mathcal{Y}_t, \mathcal{A})_{(j)} \text{Reshape}(\mathcal{Y}_t, \mathcal{A})'_{(j)}$$

on the main diagonal, such that $\tilde{\mathbf{D}}_j = \tilde{\mathbf{Q}}'_j (T^{-1} \sum_{t=1}^T \text{Reshape}(\mathcal{Y}_t, \mathcal{A})_{(j)} \text{Reshape}(\mathcal{Y}_t, \mathcal{A})'_{(j)}) \tilde{\mathbf{Q}}_j$. Similarly, $\tilde{\mathbf{D}}_V$ denotes the $r_V \times r_V$ diagonal matrix with the first largest r_V eigenvalues of

$$\frac{1}{T} \sum_{t=1}^T \text{Reshape}(\mathcal{Y}_t, \mathcal{A})_{(v)} \text{Reshape}(\mathcal{Y}_t, \mathcal{A})'_{(v)}.$$

Correspondingly, define an $r_j \times r_j$ matrix and an $r_V \times r_V$ matrix that

$$\begin{aligned} \tilde{\mathbf{H}}_j &:= T^{-1} \tilde{\mathbf{D}}_j^{-1} \tilde{\mathbf{Q}}'_j \mathbf{Q}_j \sum_{t=1}^T \mathbf{F}_{\text{reshape}, Z, t, (j)} \mathbf{F}'_{\text{reshape}, Z, t, (j)}, \\ \tilde{\mathbf{H}}_V &:= T^{-1} \tilde{\mathbf{D}}_V^{-1} \tilde{\mathbf{Q}}'_V \mathbf{Q}_V \sum_{t=1}^T \mathbf{F}_{\text{reshape}, Z, t, (v)} \mathbf{F}'_{\text{reshape}, Z, t, (v)}, \end{aligned}$$

where $\mathcal{F}_{\text{reshape}, Z, t} := \mathcal{F}_{\text{reshape}, t} \times_{j=1}^{v-1} \mathbf{Z}_j^{1/2} \times_v \mathbf{Z}_V^{1/2}$. As $T, d_1, \dots, d_{v-1}, d_V \rightarrow \infty$ we have $\{\tilde{\mathbf{H}}_j\}_{j \in [v-1]}, \tilde{\mathbf{H}}_V$ are invertible, and for $j \in [v-1]$ that $\|\tilde{\mathbf{H}}_j\|_F = O_P(1)$ and $\|\tilde{\mathbf{H}}_V\|_F = O_P(1)$. For each $j \in [v-1]$,

$$\|\tilde{\mathbf{Q}}_j - \mathbf{Q}_j \tilde{\mathbf{H}}'_j\|_F^2 = O_P \left\{ d_j^{2(\delta_{j,1} - \delta_{j,r_j})} \left(\frac{1}{T d_{\cdot k} d_j^{1 - \delta_{j,1}}} + \frac{1}{d_j^{1 + \delta_{j,r_j}}} \right) \frac{d^2}{\gamma_s^2} \right\}, \quad (\text{S2.28})$$

$$\|\tilde{\mathbf{Q}}_V - \mathbf{Q}_V \tilde{\mathbf{H}}'_V\|_F^2 = O_P \left\{ d_V^{2(\delta_{V,1} - \delta_{V,r_V})} \left(\frac{1}{T d d_V^{-\delta_{V,1}}} + \frac{1}{d_V^{1 + \delta_{V,r_V}}} \right) \frac{d^2}{\gamma_s^2} \right\}, \quad (\text{S2.29})$$

where γ_s is defined in Assumption (R1). Lastly, the i -th entry of the vectorised common component in (3.3) is also consistent such that

$$\begin{aligned} & \left\{ (\mathbf{vec}(\tilde{\mathcal{C}}_{\text{reshape}, t}))_i - (\mathbf{vec}(\mathcal{C}_{\text{reshape}, t}))_i \right\}^2 \\ &= O_P \left(\max_{j \in [v-1]} \left\{ d_j^{2(\delta_{j,1} - \delta_{j,r_j})} \left(\frac{1}{T d_{\cdot k} d_j^{1 - \delta_{j,1}}} + \frac{1}{d_j^{1 + \delta_{j,r_j}}} \right) \frac{d^2}{\gamma_s \gamma_w} \right\} \right. \\ & \quad \left. + d_V^{2(\delta_{V,1} - \delta_{V,r_V})} \left(\frac{1}{T d d_V^{-\delta_{V,1}}} + \frac{1}{d_V^{1 + \delta_{V,r_V}}} \right) \frac{d^2}{\gamma_s \gamma_w} + \frac{d}{\gamma_w^2} \right), \end{aligned} \quad (\text{S2.30})$$

where γ_w is defined in Assumption (R2).

Proof of Lemma 7. Consider (4.7), we have by Assumptions (E1) and (E2) that

$$\text{Reshape}(\mathcal{F}_{e,t}, \mathcal{A}) = \sum_{q \geq 0} a_{e,q} \text{Reshape}(\mathcal{X}_{e,t-q}, \mathcal{A}), \quad \text{Reshape}(\epsilon_t, \mathcal{A}) = \sum_{q \geq 0} a_{\epsilon,q} \text{Reshape}(\mathcal{X}_{\epsilon,t-q}, \mathcal{A}),$$

which implies that the structure depicted in (E1) and (E2) for the noise $\mathcal{E}_t$ in (3.5) holds for $\mathcal{E}_{\text{reshape},t}$ in (3.3). Read $\text{Reshape}(\mathcal{Y}_t, \mathcal{A})$ as an order- v tensor and consider the factor model (3.3). Statements (S2.28) and (S2.29) can be shown in exactly the same way as Lemma 5 (without (L2) now; details omitted here), given that all rate conditions used in the proof of Lemma 5 are fulfilled for (3.3). In other words, it remains to show the rate conditions similar to the last equality of (S2.15) are satisfied. Note that for $j \in [v-1]$, (S2.28) requires the same rate condition as Lemma 5. Hence we are left with the rate conditions for (S2.29), i.e.,

$$d\gamma_s^{-2} T^{-1} d_V^{2(\delta_{V,1} - \delta_{V,r_V})+1} = o(1), \quad d\gamma_s^{-1} d_V^{\delta_{V,1} - \delta_{V,r_V} - 1/2} = o(1),$$

which are included in Assumption (R1). With (S2.28) and (S2.29) shown, (S2.30) follows similarly as Lemma 6 (omitted). The proof of Lemma 7 is then complete. $\square$

S2.2 Proof of theorems

Proof of Theorem 1. Let $\mathcal{A} = \{a_1, \dots, a_\ell\}$ be given. We first show that the Tucker-decomposition tensor factor model (3.2) implies (3.1) with $\mathbf{A}_{\text{reshape}, K-\ell+1} \in \mathcal{K}_{d_{a_1} \times \dots \times d_{a_\ell}}$. Suppose $\mathcal{Y}_t = \mathcal{F}_t \times_{k=1}^K \mathbf{A}_k + \mathcal{E}_t$.

Consider first $\ell = 1$. With any $\mathcal{A} = \{a_1\}$ (hence the corresponding $\mathcal{A}^* = [K] \setminus \mathcal{A}$ as defined in Theorem 1), it is direct that

$$\begin{aligned} \text{Reshape}(\mathcal{Y}_t, \mathcal{A}) &= \text{Reshape}(\mathcal{F}_t \times_{k=1}^K \mathbf{A}_k + \mathcal{E}_t, \mathcal{A}) = \text{Reshape}(\mathcal{F}_t \times_{k=1}^K \mathbf{A}_k, \{a_1\}) + \text{Reshape}(\mathcal{E}_t, \mathcal{A}) \\ &= \text{Fold}_K(\text{mat}_{a_1}(\mathcal{F}_t \times_{k=1}^K \mathbf{A}_k), \{d_1, \dots, d_{a_1-1}, d_{a_1+1}, \dots, d_K, d_{a_1}\}) + \text{Reshape}(\mathcal{E}_t, \mathcal{A}) \\ &= \text{Fold}_K(\mathbf{A}_{a_1} \text{mat}_{a_1}(\mathcal{F}_t) \mathbf{A}'_{-a_1}, \{d_1, \dots, d_{a_1-1}, d_{a_1+1}, \dots, d_K, d_{a_1}\}) + \text{Reshape}(\mathcal{E}_t, \mathcal{A}), \\ &= \text{Reshape}(\mathcal{F}_t, \mathcal{A}) \times_{i=1}^{K-1} \mathbf{A}_{\mathcal{A}_i^*} \times_K \mathbf{A}_{a_1} + \text{Reshape}(\mathcal{E}_t, \mathcal{A}), \end{aligned}$$

where the last equality used the fact that

$$\begin{aligned} \text{mat}_K \left\{ \text{Reshape}(\mathcal{F}_t, \mathcal{A}) \times_{i=1}^{K-1} \mathbf{A}_{\mathcal{A}_i^*} \times_K \mathbf{A}_{a_1} \right\} &= \mathbf{A}_{a_1} \text{mat}_K \left\{ \text{Reshape}(\mathcal{F}_t, \{a_1\}) \right\} \mathbf{A}'_{-a_1} \\ &= \mathbf{A}_{a_1} \text{mat}_K \left\{ \text{Fold}_K(\text{mat}_{a_1}(\mathcal{F}_t), \{d_1, \dots, d_{a_1-1}, d_{a_1+1}, \dots, d_K, d_{a_1}\}) \right\} \mathbf{A}'_{-a_1} = \mathbf{A}_{a_1} \text{mat}_{a_1}(\mathcal{F}_t) \mathbf{A}'_{-a_1}. \end{aligned} \tag{S2.31}$$

Hence, $\mathcal{Y}_t$ follows (3.1) with variables defined according to Theorem 1, and the model has a Kronecker structure product since $\mathbf{A}_{a_1} \in \mathcal{K}_{d_{a_1}}$.

Consider $\ell = 2$ (implying at least $K = 2$). In the following, we use the neater notation that mode- k unfolding of some tensor $\mathcal{X}$ is $\mathbf{X}_{(k)}$. Without loss of generality, let $\mathcal{A} = \{a, b\}$ (with $a < b$) and the corresponding $\mathcal{A}^* = [K] \setminus \mathcal{A}$. Take the mode- b unfolding on each $\mathcal{Y}_t$, we obtain

$$\mathbf{Y}_{t,(b)} = \mathbf{A}_b \mathbf{F}_{t,(b)} (\mathbf{A}_K \otimes \cdots \otimes \mathbf{A}_{b+1} \otimes \mathbf{A}_{b-1} \otimes \cdots \otimes \mathbf{A}_a \otimes \cdots \otimes \mathbf{A}_1)' + \mathbf{E}_{t,(b)},$$

then for each row (as a column vector) of $\mathbf{Y}_{t,(b)}$, we fold it back to an order- $(K-1)$ tensor along the remaining dimensions, i.e., for any i -th row $\mathbf{Y}_{t,(b),i}$ with $i \in [d_b]$,

$$\begin{aligned} \mathcal{Y}_{t,(b),i} &:= \text{Fold}(\mathbf{Y}_{t,(b),i}, \{d_1, \dots, d_{b-1}, d_{b+1}, \dots, d_K\}) \\ &= \text{Fold}((\mathbf{A}_K \otimes \cdots \otimes \mathbf{A}_{b+1} \otimes \mathbf{A}_{b-1} \otimes \cdots \otimes \mathbf{A}_a \otimes \cdots \otimes \mathbf{A}_1) \mathbf{F}_{t,(b)}' \mathbf{A}_{b,i}, \{d_1, \dots, d_{b-1}, d_{b+1}, \dots, d_K\}) \\ &\quad + \text{Fold}(\mathbf{E}_{t,(b),i}, \{d_1, \dots, d_{b-1}, d_{b+1}, \dots, d_K\}) \\ &= \text{Fold}(\mathbf{F}_{t,(b)}' \mathbf{A}_{b,i}, \{r_1, \dots, r_{b-1}, r_{b+1}, \dots, r_K\}) \times_{k=1}^{b-1} \mathbf{A}_k \times_{h=b}^{K-1} \mathbf{A}_{h+1} \\ &\quad + \text{Fold}(\mathbf{E}_{t,(b),i}, \{d_1, \dots, d_{b-1}, d_{b+1}, \dots, d_K\}). \end{aligned}$$

Define $\mathbf{A}_{-b,-a} := \mathbf{A}_K \otimes \cdots \otimes \mathbf{A}_{b+1} \otimes \mathbf{A}_{b-1} \otimes \cdots \otimes \mathbf{A}_{a+1} \otimes \mathbf{A}_{a-1} \otimes \cdots \otimes \mathbf{A}_1$, where by convention $\mathbf{A}_{-b,-a} = 1$ if $K = 2$. Take the mode- a unfolding on $\mathcal{Y}_{t,(b),i}$, we have

$$\begin{aligned} &(\mathcal{Y}_{t,(b),i})_{(a)} \\ &= \mathbf{A}_a \left\{ \text{Fold}(\mathbf{F}_{t,(b)}' \mathbf{A}_{b,i}, \{r_1, \dots, r_{a-1}, r_{a+1}, \dots, r_K\}) \right\}_{(a)} \mathbf{A}_{-b,-a}' \\ &\quad + \left\{ \text{Fold}(\mathbf{E}_{t,(b),i}, \{d_1, \dots, d_{b-1}, d_{b+1}, \dots, d_K\}) \right\}_{(a)} \\ &= \sum_{j=1}^{r_b} \mathbf{A}_a \left\{ \text{Fold}(\mathbf{A}_{b,i,j} \mathbf{F}_{t,(b),j}, \{r_1, \dots, r_{b-1}, r_{b+1}, \dots, r_K\}) \right\}_{(a)} \mathbf{A}_{-b,-a}' \\ &\quad + \left\{ \text{Fold}(\mathbf{E}_{t,(b),i}, \{d_1, \dots, d_{b-1}, d_{b+1}, \dots, d_K\}) \right\}_{(a)} \\ &= (\mathbf{A}_{b,i}' \otimes \mathbf{I}_{d_a}) (\mathbf{I}_{r_b} \otimes \mathbf{A}_a) \begin{pmatrix} \left\{ \text{Fold}(\mathbf{F}_{t,(b),1}, \{r_1, \dots, r_{b-1}, r_{b+1}, \dots, r_K\}) \right\}_{(a)} \mathbf{A}_{-b,-a}' \\ \dots \\ \left\{ \text{Fold}(\mathbf{F}_{t,(b),r_b}, \{r_1, \dots, r_{b-1}, r_{b+1}, \dots, r_K\}) \right\}_{(a)} \mathbf{A}_{-b,-a}' \end{pmatrix} \\ &\quad + \left\{ \text{Fold}(\mathbf{E}_{t,(b),i}, \{d_1, \dots, d_{b-1}, d_{b+1}, \dots, d_K\}) \right\}_{(a)} \\ &= (\mathbf{A}_{b,i}' \otimes \mathbf{A}_a) \begin{pmatrix} \left\{ \text{Fold}(\mathbf{F}_{t,(b),1}, \{r_1, \dots, r_{b-1}, r_{b+1}, \dots, r_K\}) \right\}_{(a)} \\ \dots \\ \left\{ \text{Fold}(\mathbf{F}_{t,(b),r_b}, \{r_1, \dots, r_{b-1}, r_{b+1}, \dots, r_K\}) \right\}_{(a)} \end{pmatrix} \mathbf{A}_{-b,-a}' \\ &\quad + \left\{ \text{Fold}(\mathbf{E}_{t,(b),i}, \{d_1, \dots, d_{b-1}, d_{b+1}, \dots, d_K\}) \right\}_{(a)}. \end{aligned}$$

Therefore,

$$\begin{aligned} \mathcal{Y}_{t,a \sim b} = \begin{pmatrix} (\mathcal{Y}_{t,(b),1})_{(a)} \\ \dots \\ (\mathcal{Y}_{t,(b),d_b})_{(a)} \end{pmatrix} &= (\mathbf{A}_b \otimes \mathbf{A}_a) \begin{pmatrix} \{\text{Fold}(\mathbf{F}_{t,(b),1}, \{r_1, \dots, r_{b-1}, r_{b+1}, \dots, r_K\})\}_{(a)} \\ \dots \\ \{\text{Fold}(\mathbf{F}_{t,(b),r_b}, \{r_1, \dots, r_{b-1}, r_{b+1}, \dots, r_K\})\}_{(a)} \end{pmatrix} \mathbf{A}'_{-b,-a} \\ &+ \begin{pmatrix} \{\text{Fold}(\mathbf{E}_{t,(b),1}, \{d_1, \dots, d_{b-1}, d_{b+1}, \dots, d_K\})\}_{(a)} \\ \dots \\ \{\text{Fold}(\mathbf{E}_{t,(b),d_b}, \{d_1, \dots, d_{b-1}, d_{b+1}, \dots, d_K\})\}_{(a)} \end{pmatrix}, \end{aligned}$$

so that by definition of the reshape operator with $\ell = 2$,

$$\begin{aligned} \text{Reshape}(\mathcal{Y}_t, \{a, b\}) &= \text{Fold}_{K-1}(\mathcal{Y}_{t,a \sim b}, \{d_1, \dots, d_{a-1}, d_{a+1}, \dots, d_{b-1}, d_{b+1}, \dots, d_K, d_a d_b\}) \\ &= \text{Reshape}(\mathcal{F}_t, \{a, b\}) \times_{i=1}^{K-2} \mathbf{A}_{\mathcal{A}_i^*} \times_{K-1} (\mathbf{A}_b \otimes \mathbf{A}_a) + \text{Reshape}(\mathcal{E}_t, \{a, b\}), \end{aligned}$$

where the last line used similar arguments (omitted) as (S2.31). This implies $\mathcal{Y}_t$ follows (3.1) with a Kronecker structure product as $(\mathbf{A}_b \otimes \mathbf{A}_a) \in \mathcal{K}_{d_a \times d_b}$.

Finally, consider any $\ell \geq 3$. We use an induction argument. With the definition of tensor reshape in Section 2.2 and the above for $\ell = 2$, the initial case $\ell = 3$ can be shown by

$$\begin{aligned} \text{Reshape}(\mathcal{Y}_t, \{a_1, a_2, a_3\}) &= \text{Reshape}[\text{Reshape}(\mathcal{Y}_t, \{a_2, a_3\}), \{a_1, K-1\}] \\ &= \text{Reshape}\left\{ \text{Reshape}(\mathcal{F}_t, \{a_2, a_3\}) \times_{i=1}^{K-2} \mathbf{A}_{[[K] \setminus \{a_2, a_3\}]_i} \times_{K-1} (\mathbf{A}_{a_3} \otimes \mathbf{A}_{a_2}) \right. \\ &\quad \left. + \text{Reshape}(\mathcal{E}_t, \{a_2, a_3\}), \{a_1, K-1\} \right\} \\ &= \text{Reshape}\left\{ \text{Reshape}(\mathcal{F}_t, \{a_2, a_3\}), \{a_1, K-1\} \right\} \times_{i=1}^{K-3} \mathbf{A}_{[[K] \setminus \{a_2, a_3\}] \setminus \{a_1\}]_i} \times_{K-2} (\mathbf{A}_{a_3} \otimes \mathbf{A}_{a_2} \otimes \mathbf{A}_{a_1}) \\ &\quad + \text{Reshape}\left\{ \text{Reshape}(\mathcal{E}_t, \{a_2, a_3\}), \{a_1, K-1\} \right\} \\ &= \text{Reshape}(\mathcal{F}_t, \{a_1, a_2, a_3\}) \times_{i=1}^{K-3} \mathbf{A}_{[[K] \setminus \{a_1, a_2, a_3\}]_i} \times_{K-2} (\mathbf{A}_{a_3} \otimes \mathbf{A}_{a_2} \otimes \mathbf{A}_{a_1}) \\ &\quad + \text{Reshape}(\mathcal{E}_t, \{a_1, a_2, a_3\}), \end{aligned} \tag{S2.32}$$

where the last equality used again the definition of tensor shape, and note that $(\mathbf{A}_{a_3} \otimes \mathbf{A}_{a_2} \otimes \mathbf{A}_{a_1}) \in \mathcal{K}_{d_{a_1} \times d_{a_2} \times d_{a_3}}$. Now if for all $\ell \in [L]$ with $L \geq 3$, (3.2) implies (3.1) with $\mathbf{A}_{\text{reshape}, K-\ell+1} \in \mathcal{K}_{d_{a_1} \times \dots \times d_{a_\ell}}$, which is then also true for $\ell = L+1$ in a similar argument (omitted) as (S2.32). This completes the induction.

Given any $\mathcal{A}$, note also that if Assumption (F1) holds with $\mathcal{X}_{\text{reshape}, t}$ and $\mathcal{F}_{\text{reshape}, t}$ replaced by $\mathcal{X}_t$ and $\mathcal{F}_t$ respectively (with $\mathcal{X}_t$ and $\mathcal{F}_t$ from (4.5)), then it is immediate from the linearity of the reshape operator

that

$$\mathcal{F}_{\text{reshape},t} = \text{Reshape}(\mathcal{F}_t, \mathcal{A}) = \text{Reshape}\left(\sum_{w \geq 0} a_{f,w} \mathcal{X}_{f,t-w}, \mathcal{A}\right) = \sum_{w \geq 0} a_{f,w} \text{Reshape}(\mathcal{X}_{f,t-w}, \mathcal{A}),$$

which implies $\mathcal{F}_{\text{reshape},t}$ follows Assumption (F1) by $\mathcal{X}_{\text{reshape},t} = \text{Reshape}(\mathcal{X}_{f,t}, \mathcal{A})$.

We have now proved that (3.2) uniquely implies (3.1) with $\mathbf{A}_{\text{reshape},K-\ell+1} \in \mathcal{K}_{d_{a_1} \times \dots \times d_{a_\ell}}$, and a version of Assumption (F1) on $\{\mathcal{F}_t\}$ (from (4.5)) implies Assumption (F1) on $\mathcal{F}_{\text{reshape},t}$. It remains to show the other way around (for some $\mathcal{A}$), but all the previous steps are reversible (note that in particular, the reshape operator is reversible as long as the dimension of the original tensor is known) and $\mathbf{A}_{\text{reshape},K-\ell+1} \in \mathcal{K}_{d_{a_1} \times \dots \times d_{a_\ell}}$ ensures the existence of an appropriate set of low-rank matrices. Therefore, the proof for the theorem is completed. $\square$

Proof of Theorem 2. Under H_0 , consider (3.5). Since there exists $m \in [|\mathcal{R}|]$ such that

$$(\pi_{m,1}, \dots, \pi_{m,K-v+1}) = (r_v, \dots, r_K),$$

we only consider such m and simplify $\widehat{\mathcal{C}}_{m,t}$ as $\widehat{\mathcal{C}}_t$ and $\widehat{\mathcal{E}}_{m,t}$ as $\widehat{\mathcal{E}}_t$ (see the explanation at the beginning of Section S2). By (3.10) and Assumption (E1),

$$\begin{aligned} \widehat{\mathbf{E}}_{t,(k^*)} &= \text{mat}_{k^*}(\widehat{\mathcal{E}}_t) = \text{mat}_{k^*}((\mathcal{C}_t - \widehat{\mathcal{C}}_t) + \mathcal{F}_{e,t} \times_1 \mathbf{A}_{e,1} \times_2 \dots \times_K \mathbf{A}_{e,K} + \boldsymbol{\Sigma}_\epsilon \circ \boldsymbol{\epsilon}_t) \\ &= (\mathcal{C}_t - \widehat{\mathcal{C}}_t)_{(k^*)} + \mathbf{A}_{e,k^*} \mathbf{F}_{e,t,(k^*)} \mathbf{A}'_{e,-k^*} + \boldsymbol{\Sigma}_{\epsilon,(k^*)} \circ \boldsymbol{\epsilon}_{t,(k^*)}, \end{aligned}$$

where $\mathbf{A}_{e,-k^*} := \mathbf{A}_{e,K} \otimes \dots \otimes \mathbf{A}_{e,k^*+1} \otimes \mathbf{A}_{e,k^*-1} \otimes \dots \otimes \mathbf{A}_{e,1}$. Hence, for any $t \in [T]$ and $j \in [d/d_k^*]$, we have

$$\begin{aligned} \frac{1}{d_{k^*}} \sum_{i=1}^{d_{k^*}} (\widehat{E}_{t,(k^*),ij}^2 - \Sigma_{\epsilon,(k^*),ij}^2) &= \frac{1}{d_{k^*}} (\widehat{\mathbf{E}}'_{t,(k^*)} \widehat{\mathbf{E}}_{t,(k^*)})_{jj} - \frac{1}{d_{k^*}} \sum_{i=1}^{d_{k^*}} \Sigma_{\epsilon,(k^*),ij}^2 = \sum_{h=1}^6 \mathcal{I}_{e,h}, \\ \text{where } \mathcal{I}_{e,1} &:= \frac{1}{d_{k^*}} \{(\boldsymbol{\Sigma}_{\epsilon,(k^*)} \circ \boldsymbol{\epsilon}_{t,(k^*)})' (\boldsymbol{\Sigma}_{\epsilon,(k^*)} \circ \boldsymbol{\epsilon}_{t,(k^*)})\}_{jj} - \frac{1}{d_{k^*}} \sum_{i=1}^{d_{k^*}} \Sigma_{\epsilon,(k^*),ij}^2, \\ \mathcal{I}_{e,2} &:= \frac{1}{d_{k^*}} \{(\mathcal{C}_t - \widehat{\mathcal{C}}_t)'_{(k^*)} (\mathcal{C}_t - \widehat{\mathcal{C}}_t)_{(k^*)}\}_{jj}, \\ \mathcal{I}_{e,3} &:= \frac{1}{d_{k^*}} \{\mathbf{A}_{e,-k^*} \mathbf{F}'_{e,t,(k^*)} \mathbf{A}'_{e,k^*} \mathbf{A}_{e,k^*} \mathbf{F}_{e,t,(k^*)} \mathbf{A}'_{e,-k^*}\}_{jj}, \\ \mathcal{I}_{e,4} &:= O_P\left(d_{k^*}^{-1} \{(\mathcal{C}_t - \widehat{\mathcal{C}}_t)'_{(k^*)} \mathbf{A}_{e,k^*} \mathbf{F}_{e,t,(k^*)} \mathbf{A}'_{e,-k^*}\}_{jj}\right), \\ \mathcal{I}_{e,5} &:= O_P\left(d_{k^*}^{-1} \{(\mathcal{C}_t - \widehat{\mathcal{C}}_t)'_{(k^*)} (\boldsymbol{\Sigma}_{\epsilon,(k^*)} \circ \boldsymbol{\epsilon}_{t,(k^*)})\}_{jj}\right), \\ \mathcal{I}_{e,6} &:= O_P\left(d_{k^*}^{-1} \{\mathbf{A}_{e,-k^*} \mathbf{F}'_{e,t,(k^*)} \mathbf{A}'_{e,k^*} (\boldsymbol{\Sigma}_{\epsilon,(k^*)} \circ \boldsymbol{\epsilon}_{t,(k^*)})\}_{jj}\right). \end{aligned} \tag{S2.33}$$

Consider $\mathcal{I}_{e,2}$. From Lemma 6, recall that

$$\begin{aligned} & \left\{ (\mathbf{vec}(\widehat{\mathcal{C}}_t))_i - (\mathbf{vec}(\mathcal{C}_t))_i \right\}^2 \\ &= O_P \left(\max_{k \in [K]} \left\{ d_k^{2(\delta_{k,1} - \delta_{k,r_k})} \left(\frac{1}{T d_{-k} d_k^{1-\delta_{k,1}}} + \frac{1}{d_k^{1+\delta_{k,r_k}}} \right) \frac{d^2}{g_s g_w} \right\} + \frac{d}{g_w^2} \right), \end{aligned}$$

which is the squared error for each entry of $\widehat{\mathcal{C}}_t$. With Assumption (R2), the above squared error rate is $o(1/\min_{k \in [K]} \{d_k\}) = o(d_{k^*}^{-1})$. Hence $\mathcal{I}_{e,2} = o_P(d_{k^*}^{-1})$.

With Assumption (E1), $\|\mathbf{A}_{e,k^*}\|_F = O(1)$, $\|\mathbf{A}_{e,-k^*}\|_F = O(1)$ and $r_{e,k}$ for $k \in [K]$ are finite, so that $\mathcal{I}_{e,3}, \mathcal{I}_{e,6} = O_P(d_{k^*}^{-1})$. By the Cauchy–Schwarz inequality, immediately $\mathcal{I}_{e,4} = O_P(\mathcal{I}_{e,2}^{1/2} \cdot \mathcal{I}_{e,3}^{1/2}) = o_P(d_{k^*}^{-1})$.

Consider $\mathcal{I}_{e,1}$, noting that

$$\mathcal{I}_{e,1} = \frac{1}{d_{k^*}} \sum_{i=1}^{d_{k^*}} \Sigma_{\epsilon, (k^*), ij}^2 (\epsilon_{t, (k^*), ij}^2 - 1),$$

so that with Theorem 1 in Ayvazyan and Ulyanov (2023), we have

$$Z_j := \frac{d_{k^*}^{-1} \sum_{i=1}^{d_{k^*}} \Sigma_{\epsilon, (k^*), ij}^2 (\epsilon_{t, (k^*), ij}^2 - 1)}{\sqrt{d_{k^*}^{-2} \sum_{i=1}^{d_{k^*}} \text{Var}(\epsilon_{t, (k^*), ij}^2) \Sigma_{\epsilon, (k^*), ij}^4}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1),$$

implying $\mathcal{I}_{e,1}$ is of the rate $d_{k^*}^{-1/2}$ exactly. Note also that Z_j 's are independent of each other by Assumption (E1). It also follows that $\mathcal{I}_{e,5} = O_P(\mathcal{I}_{e,2}^{1/2} \cdot \mathcal{I}_{e,1}^{1/2}) = o_P(d_{k^*}^{-3/4})$. Finally, with (S2.33),

$$\begin{aligned} \frac{\sum_{i=1}^{d_{k^*}} (\widehat{E}_{t, (k^*), ij}^2 - \Sigma_{\epsilon, (k^*), ij}^2)}{\sqrt{\sum_{i=1}^{d_{k^*}} \text{Var}(\epsilon_{t, (k^*), ij}^2) \Sigma_{\epsilon, (k^*), ij}^4}} &= \frac{d_{k^*}^{-1} \sum_{i=1}^{d_{k^*}} (\widehat{E}_{t, (k^*), ij}^2 - \Sigma_{\epsilon, (k^*), ij}^2)}{\sqrt{d_{k^*}^{-2} \sum_{i=1}^{d_{k^*}} \text{Var}(\epsilon_{t, (k^*), ij}^2) \Sigma_{\epsilon, (k^*), ij}^4}} \\ &= Z_j (1 + o_P(1)) \xrightarrow{P} Z_j \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1). \end{aligned}$$

This shows the asymptotic result for $\widehat{\mathcal{E}}_t$ in Theorem 2 (i.e., the first asymptotic result). The result for $\widetilde{\mathcal{E}}_t$ can be shown in the same manner, except that for $\mathcal{I}_{e,2}$, we use (R2) on the squared error for each entry of $\widetilde{\mathcal{C}}_{\text{reshape}, t}$ which is the following from Lemma 7,

$$\begin{aligned} & \left\{ (\mathbf{vec}(\widetilde{\mathcal{C}}_{\text{reshape}, t}))_i - (\mathbf{vec}(\mathcal{C}_{\text{reshape}, t}))_i \right\}^2 \\ &= O_P \left(\max_{j \in [v-1]} \left\{ d_j^{2(\delta_{j,1} - \delta_{j,r_j})} \left(\frac{1}{T d_{-k} d_j^{1-\delta_{j,1}}} + \frac{1}{d_j^{1+\delta_{j,r_j}}} \right) \frac{d^2}{\gamma_s \gamma_w} \right\} \right. \\ & \quad \left. + d_V^{2(\delta_{V,1} - \delta_{V,r_V})} \left(\frac{1}{T d d_V^{1-\delta_{V,1}}} + \frac{1}{d_V^{1+\delta_{V,r_V}}} \right) \frac{d^2}{\gamma_s \gamma_w} + \frac{d}{\gamma_w^2} \right). \end{aligned}$$

Under H_0 , we arrive at the same conclusion with the same Z_j 's. Moreover, the asymptotic result for $\hat{\mathcal{E}}_t$ holds true under H_1 . This concludes the proof of the theorem. $\square$

Proof of Theorem 3. Under H_0 , Theorem 2 implies that for each $t \in [T]$ and $j \in [d/d_{k^*}]$, there exists such $\sum_{i=1}^{d_{k^*}} \hat{E}_{m,t,(k^*),ij}^2$ distributed asymptotically the same as $\sum_{i=1}^{d_{k^*}} \tilde{E}_{t,(k^*),ij}^2$. With the definition $\hat{q}_{y,m,j}(\alpha) := \inf \{c \mid \mathbb{P}_{y,m,j}(c) \geq 1 - \alpha\}$, we arrive at the existence of such $\hat{q}_{y,m,j}(\alpha)$ asymptotically the same as $\hat{q}_{x,j}(\alpha)$. In this case, we have

$$\alpha \geq \mathbb{P}_{y,m,j}(y_{m,j,t} \geq \hat{q}_{y,m,j}(\alpha)) \rightarrow \mathbb{P}_{y,m,j}(y_{m,j,t} \geq \hat{q}_{x,j}(\alpha)).$$

With the above, (4.8) is direct from the definition of $\mathbb{P}_{y,m,j}$. This completes the proof of the theorem. $\square$

Proof of Theorem 4. Let $\mathcal{Y}_t \in \mathbb{R}^{d_1 \times \dots \times d_K}$ be an order- K tensor and $\mathcal{A} = \{a_1, \dots, a_\ell\}$. Then each $\text{Reshape}(\mathcal{Y}_t, \mathcal{A})$ is an order- $(K - \ell + 1)$ tensor. If $\{\text{Reshape}(\mathcal{Y}_t, \mathcal{A})\}$ has a Kronecker product structure, then Theorem 1 allows us to write for $t \in [T]$,

$$\text{Reshape}(\mathcal{Y}_t, \mathcal{A}) = \mathcal{F}_{\text{reshape},t} \times_{j=1}^{K-\ell} \mathbf{A}_{\text{reshape},j} \times_{K-\ell+1} \mathbf{A}_{\text{reshape},K-\ell+1} + \mathcal{E}_{\text{reshape},t}, \quad (\text{S2.34})$$

for some core factor $\{\mathcal{F}_{\text{reshape},t}\}$, loading matrices $\{\mathbf{A}_{\text{reshape},j}\}_{j \in [K-\ell+1]}$, and noise $\mathcal{E}_{\text{reshape},t}$. Immediately by Definition 2, if $\mathbf{A}_{\text{reshape},K-\ell+1} \in \mathcal{K}_{d_{a_1} \times \dots \times d_{a_\ell}}$, then $\mathcal{Y}_t$ has a Kronecker product structure; otherwise, $\{\mathcal{Y}_t\}$ has no Kronecker product structure along $\mathcal{A}$.

It remains to show that if $\{\mathcal{Y}_t\}$ either has a Kronecker product structure or has no Kronecker product structure along some set $\mathcal{A}^*$, then $\{\text{Reshape}(\mathcal{Y}_t, \mathcal{A})\}$ with $\mathcal{A}^* \subseteq \mathcal{A}$ has a Kronecker product structure. The case where $\{\mathcal{Y}_t\}$ has a Kronecker product structure is trivial. We then only need to consider that $\{\mathcal{Y}_t\}$ has no Kronecker product structure along $\mathcal{A}^*$, which implicitly assume a factor model of $\{\mathcal{Y}_t\}$ along $\mathcal{A}^*$ by Definition 2. Without loss of generality, let $\mathcal{A}^* := \{K - g + 1, \dots, K\}$, otherwise redefine the mode indices of $\mathcal{Y}_t$. For the set $\mathcal{A}$ with $\mathcal{A}^* \subseteq \mathcal{A}$, we now read $\mathcal{A} = \{a_1, \dots, a_{\ell-g}, K - g + 1, \dots, K\}$. Using the last property of tensor reshape in Section 2.2 (which can be easily seen by induction), we have

$$\text{Reshape}(\mathcal{Y}_t, \mathcal{A}) = \text{Reshape}\{\text{Reshape}(\mathcal{Y}_t, \mathcal{A}^*), \{a_1, \dots, a_{\ell-g}, K - g + 1\}\}. \quad (\text{S2.35})$$

Now that $\{\mathcal{Y}_t\}$ has no Kronecker product structure along $\mathcal{A}^*$, similar to the form (S2.34), we have for $t \in [T]$,

$$\text{Reshape}(\mathcal{Y}_t, \mathcal{A}^*) = \mathcal{F}_{\text{reshape},t}^* \times_{j=1}^{K-g} \mathbf{A}_{\text{reshape},j}^* \times_{K-g+1} \mathbf{A}_{\text{reshape},K-g+1}^* + \mathcal{E}_{\text{reshape},t}^*,$$

which implies the time series $\{\text{Reshape}(\mathcal{Y}_t, \mathcal{A}^*)\}$ follows a Tucker-decomposition TFM. According to Theorem 1, $\{\text{Reshape}(\mathcal{Y}_t, \mathcal{A}^*)\}$ follows a factor model along any index set of $\{\text{Reshape}(\mathcal{Y}_t, \mathcal{A}^*)\}$ (with Kronecker

product structure; but the factor model form is sufficient for our claim). In particular, with the index set $\{a_1, \dots, a_{\ell-g}, K - g + 1\}$, we conclude from (S2.35) that $\{\text{Reshape}(\mathcal{Y}_t, \mathcal{A})\}$ has a Kronecker product structure. This also completes the proof of the theorem. $\square$

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